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Multi sensing in metal-ceramic-polymer composites: Sensing mechanisms, synergistic effects, and applications

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Highlights

- Electro-mechanical coupling at metal–ceramic–polymer interfaces enables synergistic interaction under stimuli.
- Bio-inspired gradient interfaces (nacre/bone) facilitate hierarchical cross-scale assembly.
- Multimodal sensing mechanisms (chemical/physical) are comprehensively analyzed.
- Versatile applications demonstrated in detection, biomonitoring, human–machine interfaces, and EMI shielding.

Outline

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Abstract

Metal-ceramic-polymer multicomponent composites exhibit unique advantages in the field of physicochemical multimodal sensing due to their component designability and functional synergy. The multimodal sensing mechanism of these composites is based on their unique material combinations and interfacial interactions, and the response to external stimuli is realized through different chemical effects (chemisorption, conductivity change, surface chemical reaction, etc.) and physical effects (e.g., piezoelectric effect, photoelectric effect, electromagnetic response, etc.). During multimodal sensing, the interaction between metals, polymers, and ceramics generates robust synergies that significantly boost material performance and sensitivity. This synergy extends beyond the mere complementarity of physical properties to include the reciprocal amplification of distinct sensing mechanisms. In this paper, a systematic review of the construction strategy of this material system, the physical/chemical multimodal sensing mechanism and its application progress in complex environmental monitoring is presented. In addition, the application areas of metal-ceramic-polymer multicomponent composites are summarized, including gas/liquid sensing, human health, human-computer interaction and electromagnetic shielding.

Graphical abstract

A systematic review of the construction strategy of metal-ceramic-polymer composites, the physical/chemical multimodal sensing mechanism and its application.

5. Key applications

6. Conclusion and outlook

CRediT authorship contribution statement

Declaration of competing interest

Acknowledgements

Data availability

References

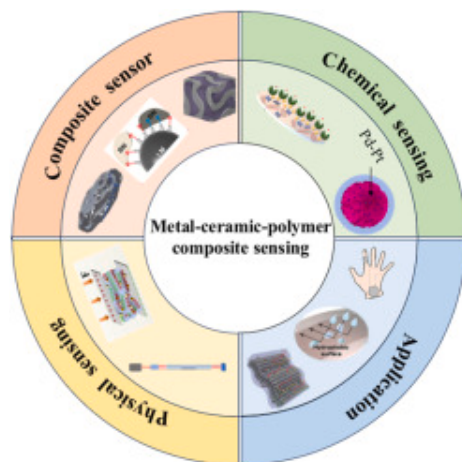
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Keywords

Metal-ceramic-polymer composites; Multimodal sensing; Synergistic effect; Chemical sensing mechanism; Physical sensing mechanism

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1. Introduction

With the rapid development of science and technology, sensor technology has been widely used in many fields such as environmental monitoring [[1], [2], [3]], medical diagnosis [4], industrial control [5] and so on. In recent years, metal-ceramic-polymer composites have received increasing attention due to their excellent physical and chemical properties [6]. These composites combine the electrical conductivity of metals, the high temperature resistance of ceramics, and the flexibility of polymers, exhibiting

good mechanical, thermal, and electrical properties, making them an important potential for sensor applications.

Metallic materials typically offer excellent electrical and thermal conductivity, making them ideal for sensors requiring high sensitivity and rapid response. Conversely, ceramic materials are favored for extreme environments due to their superior corrosion resistance and thermal stability, while polymers are valued for their processability and lightweight nature [[7], [8], [9]].

However, each single material has limitations. Metals may be deficient in terms of chemical stability and biocompatibility; polymers are relatively weak in terms of mechanical strength and thermal stability; and ceramics tend to be brittle and difficult to process. In order to overcome these limitations and integrate the advantages of different materials, metal/polymer/ceramic composites have been developed. Such composites organically combine the electrical and thermal properties of metals, the molecular recognition and flexibility of polymers, and the stability and corrosion resistance of ceramics, providing a broad scope for the development of new high-performance sensors.

In recent years, research on sensing in metal-polymer-ceramic composites has gradually become a hot topic. By carefully designing the composition, structure and preparation process of the composites, their physicochemical properties can be regulated, thus realising the sensing detection of a wide range of physical quantities (e.g., stress, [10,11], strain [[12], [13], [14]], temperature, shidu [15,16], field, etc.) and chemical substances (e.g., gases [17], ions, biomolecules, etc.) with high sensitivity, high selectivity and fast response [[18], [19], [20]]. An in-depth study of the sensing modes of such composites will not only help to promote the development of the sensing technology itself, but also have a profound impact on the related cross-disciplinary fields, and provide innovative solutions to the sensing challenges in practical engineering and scientific research.

Although numerous reviews have summarized progress in flexible electronics based on liquid metals, conductive polymers, or binary metal-oxide nanostructures, these works predominantly focus on binary synergies. However, a critical gap remains in

understanding the "Triadic Synergy" within Metal-Ceramic-Polymer (MCP) systems. Unlike binary systems that often face trade-offs between mechanical compliance and environmental stability, MCP composites leverage the third phase to resolve these conflicts—using ceramics to enhance thermal/chemical stability without sacrificing the flexibility provided by polymers or the conductivity of metals. To date, a systematic review specifically addressing the structural design and multimodal enabling mechanisms of these triadic composites is lacking.

In this review, we introduce the electro-mechanical coupling effect at metal–ceramic–polymer interfaces-highlighting synergistic interactions of conductive networks, piezoelectric ceramics, and flexible polymers under stress-chemical fields-and introduce a biomineralization-inspired graded assembly for cross-scale ordering. We also detail dual chemical-physical sensing mechanisms and review applications in gas/liquid detection, healthcare, human-machine interfaces, and electromagnetic shielding ([Fig. 1](#)).



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Fig. 1. Sensing properties and applications of metal-polymer-ceramic composites.

2. Synergistic driven metal-ceramic-polymer composites: construction strategies

The development of high-performance sensors relies on advanced construction strategies that go beyond simple blending. By engineering the architecture of metal, ceramic, and polymer phases, researchers can tailor the distribution of conductive pathways, mechanical supports, and active sensing sites.

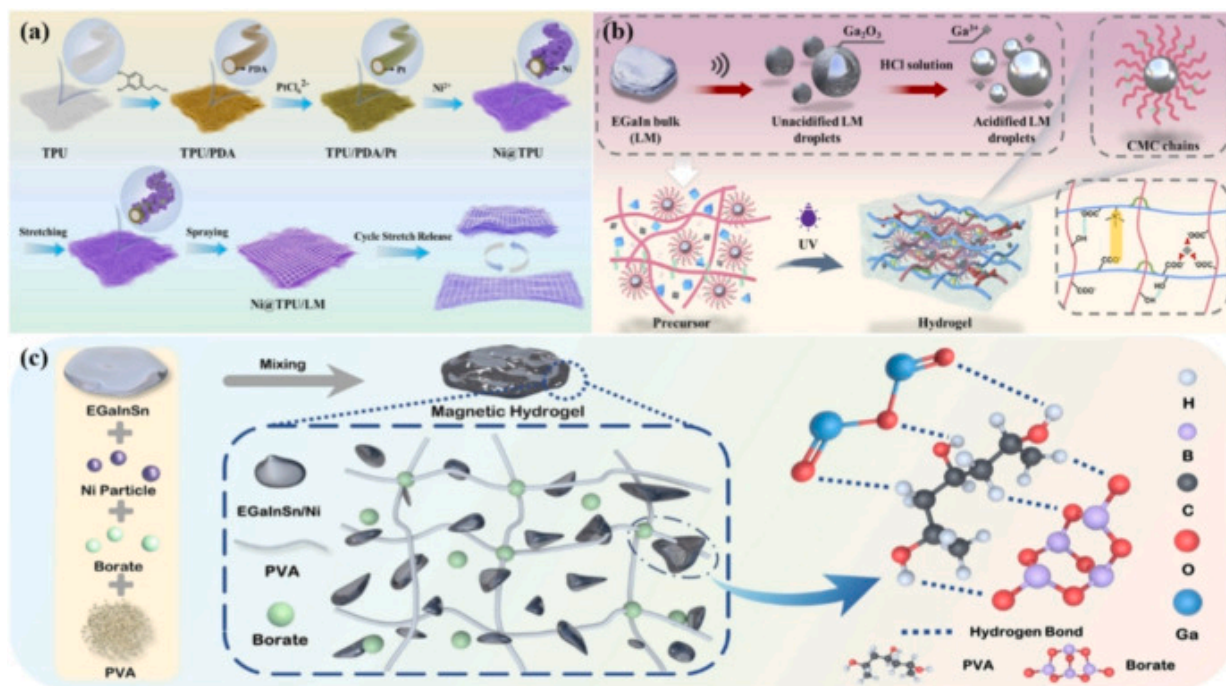
2.1. Construction of compliant conductive networks: The conductivity-flexibility synergy

The fundamental synergy in metal-polymer composites lies in resolving the intrinsic conflict between the superior electron transport of metals and the mechanical compliance of polymers. While metals typically offer high conductivity but suffer from rigidity and brittleness [21,22], polymers provide essential stretchability and corrosion resistance yet lack electrical functionality [23]. Consequently, the primary design principle involves constructing a robust percolating conductive network within a viscoelastic matrix. In this synergistic system, the polymer phase does not merely serve as a passive binder; it acts as a stress-buffering medium that accommodates significant deformation, preventing the fracture of the conductive pathways formed by metal fillers [24].

The most vivid demonstration of this "conductivity-flexibility" synergy is found in Gallium-based liquid metal (LM) elastomers, where the fluid nature of the filler perfectly complements the elasticity of the matrix. Unlike rigid solid fillers that may detach under strain, LMs possess infinite deformability, allowing conductive pathways to reconfigure dynamically. For instance, discrete LM droplets can be uniformly dispersed within a silicone elastomer matrix to create highly deformable circuits [25]. Here, the polymer matrix acts as a "viscoelastic container" that prevents LM leakage while providing the necessary restoring force after deformation, thereby maintaining electromechanical stability under cyclic loads. To further optimize this interaction, Yu et al. [26] proposed a

bio-inspired approach mimicking vascular networks by injecting LM into microchannels within a self-healing PDMS elastomer. This structural synergy ensures that the liquid conductor flows adaptively within the polymer channels during stretching, achieving a high gauge factor (3.464) without compromising structural integrity.

Extending this synergistic design to wet or biocompatible environments, researchers have developed composite hydrogels that rely on molecular-level network interactions. In these systems, Double Network (DN) strategies are often employed to balance conductivity with mechanical toughness. Yang et al. [27] fabricated an oxide-free LM-doped hydrogel by encapsulating LM droplets within a rigid polysaccharide network (CMC) while intertwining it with a soft synthetic network (Fig. 2). This interpenetrating architecture dissipates energy through the rupture of sacrificial bonds, endowing the hydrogel with exceptional stretchability (>2800%) and high conductivity (5.27 S/m). Similarly, the interface between the metal and the polymer matrix is critical for stability; Zou et al. [28] and Zhao et al. [29] utilized cellulose nanofibres (CNF) and PVA networks to stabilize LM droplets. These polymer chains effectively modify the metal interface, preventing oxidation and aggregation while bridging the significant modulus mismatch between the liquid filler and the hydrogel matrix.



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Fig. 2. (a) Schematic diagram of the fabrication of Ni@TPU/LM nonwoven fabric [34] © 2024 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies., (b) Preparation diagram of CMC/P(AA-co-DAC)/NaCl-LM hydrogel [27] © 2024 Published by Elsevier Ltd., (c) Preparation of PVA/EGaInInSn-Ni composite hydrogels containing primary crosslinked networks [29] © 2023, The Author(s).

In addition to enhancing mechanical adaptability, the polymer matrix plays a crucial role in improving the environmental stability and functional durability of metal fillers. The polymer covering serves as a barrier against oxidation and corrosion, which is particularly vital for applications in harsh environments [30,31]. This protective synergy is exemplified in electromagnetic interference (EMI) shielding materials. Ren et al. [32] reported the successful fabrication of ultrathin (70 μm), flexible, and stable EMI shielding

membranes by direct processing of composites containing liquid metals (LMs) and cellulose nanofibrils (CNFs). The membranes demonstrated an average specific EMI shielding efficiency (EMI SE) of 429 dB/mm across the 8.2–18 GHz frequency range. Xie et al. [33] fabricated and investigated ionic polymer-metal composite (IPMC) sensors, where the high flexibility and durability of IPMC is a favorable characteristic for impact sensing under harsh conditions. Lu et al. [34] fabricated a switchable and low-reflectance by chemically nickel-plating the surface of a polyurethane nonwoven fabric (Ni@TPU), which was then bonded to a liquid metal (LM) grid EMI shielding composite. In these composites, the polymer backbone provides the structural integrity to withstand bending and folding, while the continuous metal network efficiently dissipates electromagnetic waves, achieving a balance of performance that neither component could achieve in isolation.

Collectively, the evolution from rigid metal fillers to liquid metals and ionic hydrogels highlights a critical design trajectory for metal-polymer systems: matching the mechanical impedance while maintaining the percolation threshold. The most successful strategies, such as bio-inspired microchannels and double-network hydrogels, demonstrate that high performance stems not merely from mixing components but from engineering the interface geometry to decouple conductivity from strain. Consequently, future material designs should prioritize interfacial softening by utilizing liquid or semi-liquid metal phases to eliminate stress concentration at hard-soft boundaries, while simultaneously employing hierarchical constraint strategies where polymer networks spatially confine conductive fillers rather than allowing random dispersion. This approach, combined with active protection mechanisms—such as self-healing or passivating polymer chemistries—will be essential to lower the percolation threshold, enhance cyclic stability, and ensure long-term reliability in complex environments.

2.2. Construction of robust interpenetrating networks: The conductivity-stability synergy

The core design principle of metal-ceramic composites addresses the intrinsic trade-off between electrical functionality and environmental durability. While metals provide essential electron transport pathways for signal transmission, their susceptibility to oxidation, corrosion, and deformation at high temperatures limits their reliability. Conversely, ceramics act as a robust "armor," imparting exceptional thermal stability and mechanical rigidity [35]. Therefore, the primary goal of this synergy is to construct Interpenetrating Phase Composites (IPCs). In these architectures, the continuous metal phase ensures efficient charge transport, while the ceramic skeleton bears mechanical loads and resists thermal shock, creating a "Conductivity-Stability" synergy that neither component can achieve alone. Structurally, these composites typically integrate ceramic particles or fibers within a metal matrix (Fig. 3), merging the conductivity of metals with the wear resistance and thermal stability of ceramics. (e.g. high temperature, high pressure, strong corrosion, etc.).

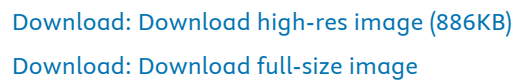


Fig. 3. Microstructure and phase composition of the metal-ceramic material. (a) TEM micrographs of the metal-ceramic material showing the ultrafine 3D interpenetrated structure with ceramic flakes preferentially aligned and alternating with the metallic phases in the natural pearl layer. (b) Ultrafine grain structure of the metallic phase and (c) separation of the ceramic flakes into smaller flakes in the composite. (d) XRD patterns of the metal-ceramic material in horizontal ($\parallel$) and vertical ($\perp$) sections, respectively [36] © 2022 Elsevier Ltd. All rights reserved. (e) A mixture of ISA terpolymer and PMVS was subjected to EISA at 40°C to produce a double gyro-structured hybrid. (f) Rendering of two interpenetrating minority networks of a dual gyro structure [37] © 2021 Wiley-VCH GmbH.

This synergistic interplay is most critical for sensors operating under extreme conditions, where the ceramic matrix effectively shields the functional metal phase. For high-temperature applications, the oxidation resistance of the ceramic phase is paramount. Schmid-Engel et al. [38] demonstrated that a Pt-SiO₂ nano-cermet system retains stable strain coefficients up to 250°C, as the silica matrix prevents platinum agglomeration and oxidation. Similarly, Zhang et al. [39] leveraged the thermal shock resistance of zirconia in a Mo-W-ZrO₂ composite to achieve rapid temperature response (<1 min) in harsh environments. Furthermore, for structural health monitoring in supersonic aircraft, Xu et al. [40] utilized a ferroelectric ceramic phase (BiFeO₃-BaTiO₃) with a high Curie temperature to maintain stable piezoelectric sensitivity alongside the metal phase at temperatures up to 200°C.

To realize this stability physically, advanced fabrication strategies such as melt infiltration and bio-inspired design are employed to optimize the geometric arrangement of the two phases. A representative strategy is the creation of hierarchical interpenetrating networks. Liu et al. [36] fabricated a nacre-like metalloceramic by infiltrating an ultrafine-grained Mg-Al-Zn alloy into a porous Ti₃AlC₂ ceramic framework. This bio-inspired architecture allows the metal to dissipate energy while the ceramic provides rigidity, significantly improving fracture toughness. In parallel, Thedford et al. [37] demonstrated that physical infiltration of molten indium into porous SiON templates creates dual

gyroscopic morphologies, proving that "soft" metals can be structurally reinforced by "hard" ceramic templates without sacrificing conductivity. Even for emerging liquid metals, Xing et al. [41] showed that doping Ga-based alloys into SiBOC ceramics creates a reactive core-shell structure, merging liquid conductivity with ceramic robustness.

Beyond mechanical and thermal reinforcement, the metal-ceramic synergy also unlocks unique functional versatility in shielding and sensing. In radiation protection, the high density of metals combined with the structural integrity of ceramics (e.g., WB/Al [42], W–B–Ni–Fe [43]) enhances neutron/gamma shielding efficiency while maintaining the mechanical strength required for structural components. Additionally, in chemical sensing, the interface between metal nanoparticles and ceramic semiconductors plays a crucial role. Liu et al. [44] enhanced the gas sensing performance of SnO₂ nanobelts by decorating them with Au nanoparticles. Here, the "spillover effect" at the metal-ceramic interface amplifies the catalytic reaction, demonstrating that the synergy extends from physical stability to chemical sensitivity.

In addition, metal-ceramic composites also show good application prospects in gas sensors, humidity sensors, force sensors, etc [45,46]. Due to their high strength, good thermoelectric properties and the ability to cope with a wide range of environmental pressures and temperature variations, these composites are able to meet the demanding requirements for sensor performance under complex working conditions. With the advancement of material science and the improvement of manufacturing process, metal-ceramic composites will play an increasingly important role in future smart sensing technology, promoting technological innovation and application expansion in related fields.

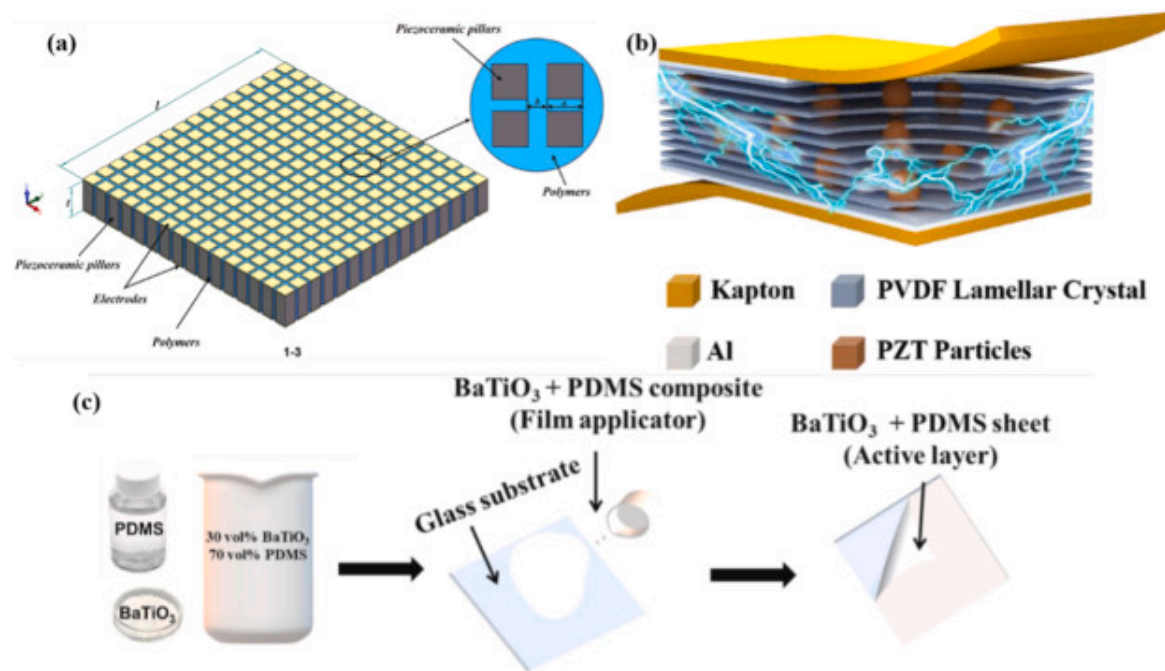
2.3. The Flexibility-Functionality synergy

Ceramics usually have high hardness, chemical stability, good thermal stability and electrical insulation [47]. Common ceramic materials include aluminium oxides (Al₂O₃), zirconium oxides (ZrO₂), and barium titanate (BaTiO₃). Ceramic materials are often used

in sensors for their stable electrical and mechanical properties, especially in temperature, pressure and strain sensing. Polymer materials, on the other hand, are malleable, flexible, and lightweight. Polymer materials are often better suited than ceramics for use in sensors that need to be flexible, lightweight, or low-cost. Commonly used polymer materials include polyvinylidene fluoride (PVDF) [48], polydimethylsiloxane (PDMS) [49], and others.

Therefore, the core objective of the "Flexibility-Functionality" synergy is to disperse functional ceramic fillers within a viscoelastic polymer matrix, ensuring that the composite retains the active response of the ceramic while adopting the deformability of the polymer [50]. To realize this synergy in strain and pressure sensors, the polymer matrix primarily serves as a stress transfer medium that protects the brittle ceramic phase during deformation. The most direct approach involves blending ceramics with ferroelectric polymers, as demonstrated by Ref. [51], which fabricated 20 μ m-thick composite films by combining PZT ceramics with PVDF. This system exhibited excellent structural stability against thermal stresses up to 160°C while delivering piezoelectric voltage coefficients comparable to bulk materials. Beyond simple blending, optimizing the geometric arrangement of the ceramic phase is crucial for enhancing electromechanical coupling. For instance, Qiu et al. [52] showed that utilizing a 1-3 connectivity structure in PZT 5A/silicone rubber composites significantly improves the coefficient (d_{33}) by reducing lateral clamping. Similarly, Chen et al. [53] leveraged this structural synergy to develop a ceramic/PDMS array for high-quality ultrasound imaging. Innovative structural designs can further amplify sensitivity; notably, Guo et al. [54] designed a "nutcracker-structured" PZT/PVDF sensor via phase separation. This unique geometry effectively concentrates stress on the piezoelectric crystals, resulting in high sensitivity (6.38mV/N) and ultra-fast response (21 ms) for real-time sports monitoring. While structural design addresses mechanical compliance, maximizing the functional output often requires high filler loading, which challenges interface compatibility. Consequently, synergistic processing strategies must focus on optimizing packing density and interfacial bonding. To overcome the trade-off between filler loading and mechanical

integrity, Fernandez et al. [55] and Sugato et al. [56] introduced an infiltration strategy by embedding small particles into the interstitial pores of larger particles in inkjet-printed films. This multi-scale synergy significantly enhanced the packing density of BaTiO₃ in PDMS, resulting in a bending sensor with high capacitance variation (42.85%) and minimal hysteresis (<2%). Ultimately, the performance is dictated by the interaction between the organic and inorganic phases. For example, Ma et al. [57] synthesized a flexible PI-b-PDMS block copolymer matrix integrated with core-shell SiO₂@BN nanofillers. The core-shell structure acts as a buffer layer to improve interfacial compatibility and thermal conductivity, raising the decomposition temperature to 472.2 °C (Fig. 4). Similarly, polydopamine (PDA) has been used as a nano-bridge to improve interfacial compatibility between BaTiO₃ nanoparticles and a PVDF matrix, mitigating agglomeration [58]. In parallel, Alzoubi et al. [59] demonstrated that tailoring the interface in Al₂O₃/PANI films allows for tunable bandgaps and high crystallinity, expanding the synergy to optoelectronic applications.



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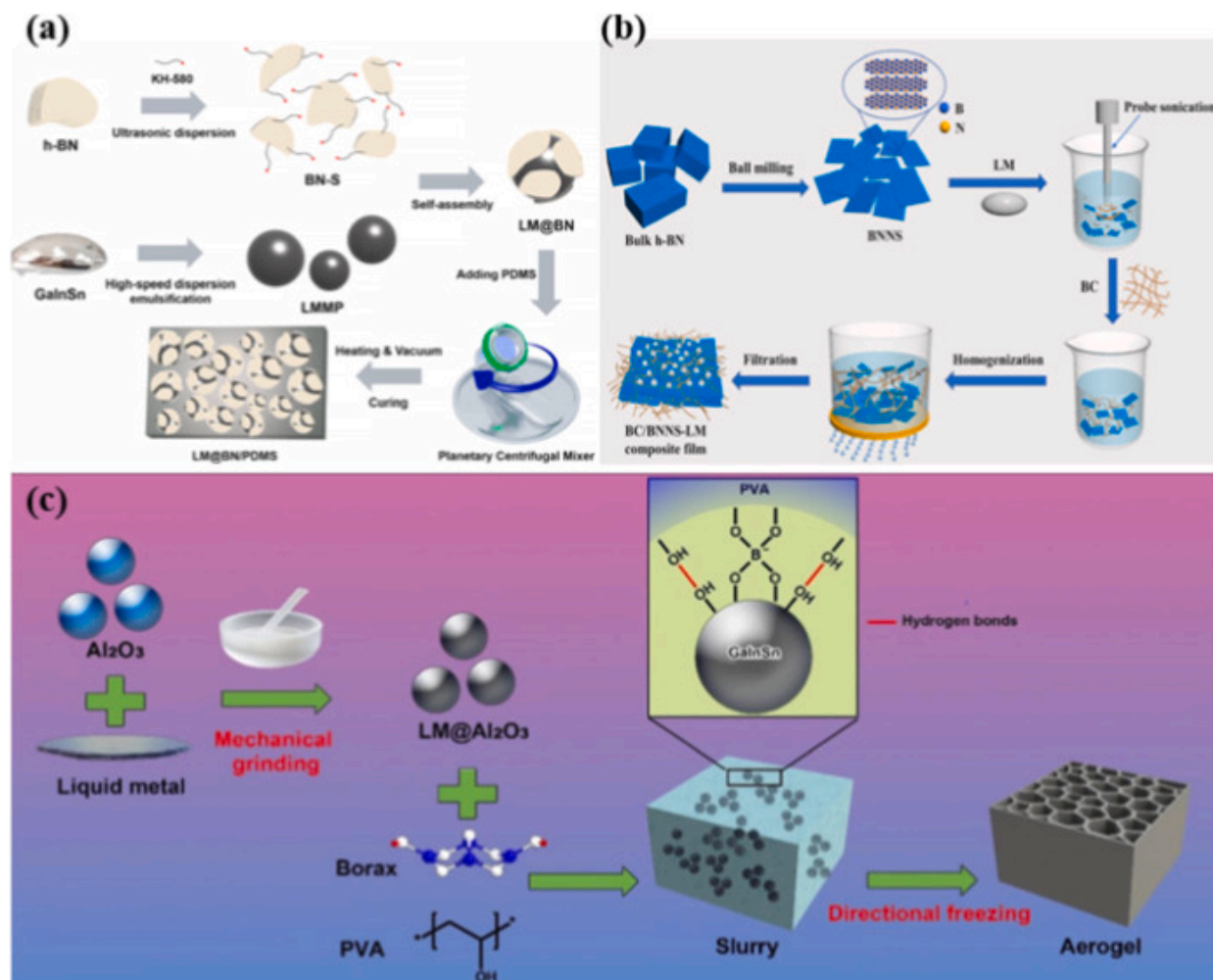
Fig. 4. (a) Structure of a piezoelectric ceramic/polymer composite array [52] © 2024 Elsevier Ltd and Techna Group S.r.l. All rights are reserved, including those for text and data mining, AI training, and similar technologies., (b) Sensor arrays fabricated from PZT/PVDF composites [54] © 2019 Elsevier Ltd. All rights reserved., (c) Preparation of BaTiO₃-PDMS composites [55] © 2024 Elsevier Ltd. All rights reserved.

2.4. Ternary collaborative construction: multi-functional integration

The construction of metal-ceramic-polymer ternary composites centers on a "Conductivity-Stability-Flexibility" triple synergy. Through advanced interfacial modification and structural design, the distinct functions of the three components are superimposed to achieve performance that exceeds the sum of the parts [60].

2.4.1. Core-shell and interfacial synergy

This strategy addresses the trade-off between thermal/electrical conductivity and mechanical insulation. Wang et al. [61] developed LM@BN core-shell structures by coating micron-sized liquid metal (LM) particles with Boron Nitride (BN) and dispersing them into an organosilicon elastomer (Fig. 5a). This design exploits the high thermal conductivity of LM, the electrical insulation of BN, and the stretchability of the polymer to achieve integrated "thermal management-insulation-flexibility." Yang et al. [62] designed the Ag@MXene-CNF (AMC) composite films, endowed with a conductive 3D architecture, are capable of furnishing an array of electromagnetic (EM) wave reflection pathways. Similarly, to reduce interfacial thermal resistance, Fan et al. [63] and Wang et al. [64] utilized an interfacial grafting strategy. By positioning modified Ga-In alloy as a bridging filler between ceramic particles (h-BN) and the PDMS matrix, they successfully bridged interfacial gaps. This synergy yielded composites with low modulus (193 kPa) and high thermal conductivity, effectively solving the "hard-soft" mismatch problem in thermal interface materials (TIMs).



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Fig. 5. (a) Schematic of the fabrication of LM@BN core-shell particles and LM@BN/PDMS composites [61] © 2023 Elsevier Ltd. All rights reserved., (b) Preparation of BC/BNNS-LM composite membrane [60] © 2024 Published by Elsevier Ltd., (c) Schematic of fabrication of Al₂O₃@LM/BCP aerogel [65] © 2025 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

2.4.2. Porous and aerogel structural synergy

For applications requiring lightweight high-strength materials, porous architectures are employed to integrate shielding and mechanical support. Chen et al. [65] fabricated an Al_2O_3 @LM/PVA aerogel via mechanical grinding and directional freeze-drying (Fig. 5c). To further enhance pressure sensitivity range, Zheng et al. [66] constructed a 3D interconnected porous TPU/MXene/LM composite foam. In this ternary system, the rigid MXene skeleton supports the porous structure, while the fluidic LM droplets trapped within the pores act as deformable capacitive plates, achieving a wide sensing range (10Pa - 260kPa) that binary foams cannot attain. Here, the synergy of the Al_2O_3 rigid skeleton, the electromagnetic shielding capability of the LM network, and the porous architecture of the PVA matrix optimized the "Shielding-Mechanics-Permeability" performance, making it ideal for flexible electronic packaging.

2.4.3. Layered and film synergy

Layered designs focus on optimizing in-plane properties and multi-parameter decoupling. Sun et al. [66] leveraged the unique film-forming capability of bacterial cellulose (BC) to immobilize BNNS and LM, fabricating flexible composite thin films. These materials exhibited exceptional in-plane thermal conductivity (66.1 W/m·K) and superior dielectric properties due to the ordered arrangement of the ceramic and metal phases within the polymer web. Recently, Fu et al. [67] advanced this strategy by constructing a heterogeneous Ag/MXene/PDMS multilayer sensor. By sequentially depositing Ag nanoparticles and MXene nanosheets onto a modified polymer substrate, they created a "dual-conductive network" where the brittle ceramic phase (MXene) induces controlled microcracks while the ductile metal phase (Ag) bridges them. This synergistic "crack-bridging" mechanism enabled precise human motion detection when coupled with deep learning algorithms. In the field of multimodal sensing, Yin et al. [68] constructed a cascaded structure combining Ag/ZnO (for Refractive Index sensing) and Ag/TiO₂/PDMS (for temperature sensing). This design integrates the plasmonic resonance of Ag, the optical response of semiconductor ceramics, and the thermo-sensitivity of PDMS,

successfully achieving simultaneous, decoupled monitoring of dual parameters. With the continuous development of material science and manufacturing technology, these triadic synergistic strategies will undoubtedly play an increasingly important role in the design of next-generation smart sensors and functional devices (Table 1).

Table 1. Overview of metal-ceramic-polymer composite sensors.

Classification	Materials	Preparation method	Machine	Appliance
Metal-polymer composites	PVA-LMP	echocardiography	strains	
Metal-polymer composites	Ni@TPU/LM	electrospinning	resistive	EMI shielding
Metal-polymer composites	CNF@LM/PAM/rGO/gelatin/glycerol	one-pot reaction	strain-stress	stress-strain sensing and temperature response
Metal-polymer composites	IPMC	conventional electroless plating	piezoelectric and pressure	advance sports science
Metal-Ceramic Composites	AuNPs/Co-doped SnO ₂	Aliovalent doping, vacuum spraying	air sensitive	gas sensor
Metal-Ceramic Composites	Mn-doped 0.67BiFeO ₃ -0.33BaTiO ₃	High temperature sintering	piezoelectric	Structural health monitoring

Classification	Materials	Preparation method	Machine	Appliance
Metal-Ceramic Composites	Pt-SiO ₂	Electrochemical deposition	strain	elevated temperature
Metal-Ceramic Composites	Tb ₂ O ₃ /ZnO	electrospinning	resistive	NO sensor
Polymer-ceramic composites	Piezoceramics/Silicon Rubber Composite Arrays	Clip Fill	piezoelectric	High-precision integrated detection
Polymer-ceramic composites	BaTiO ₃ -PDMS	inkjet printing	capacitive	human-computer interaction
Polymer-ceramic composites	PIC/SiO ₂ @BN	one-pot reaction	stress	high THz EMW transmission
Classification	Materials	Preparation method	Machine	Appliance
Polymer-ceramic composites	PANI-Al ₂ O ₃	Electrochemical deposition	optoelectronic	optical sensor

Classification	Materials	Preparation method	Machine	Appliance
Metal-ceramic-polymer composites	Ag/ZnO and Ag/TiO ₂ /PDMS	Deposition of thin films	Surface Plasmon Resonance	Fibre optics, temperature sensing
Metal-ceramic-polymer composites	Al ₂ O ₃ @LM/BCP	Mechanical grinding, melt cross-linking and directional freeze-drying techniques	-	EMI shielding
Metal-ceramic-polymer composites	Ag@MXene-CNF	vacuum filtration	stress	EMI shielding
Metal-ceramic-polymer composites	Ag/MXene/PDMS	harnessing the synergistic interplay between chemical in situ reduction and electrostatic adsorption based on the heterogeneous surface strategy	strain	wearable sensor

The progression from binary to ternary (metal-ceramic-polymer) composites represents a paradigm shift from simple property reinforcement to "Conflict Resolution" in material design. As evidenced by the core-shell (LM@BN) and aerogel architectures, the introduction of a third phase allows for the spatial segregation of conflicting functions—such as maintaining electrical insulation while maximizing thermal conductivity, or achieving electromagnetic shielding without sacrificing lightweight mechanics. Therefore, the ultimate design principle for triadic systems lies in Hierarchical Assembly: utilizing the polymer not just as a matrix, but as a "compatibilizer" to mediate the distinct physical fields of metals and ceramics. Future research should focus on precision spatial control (e.g., gradient distribution, Janus structures) to further decouple multimodal signals and minimize interfacial thermal/electrical resistance, thereby enabling truly integrated, multifunctional sensing platforms.

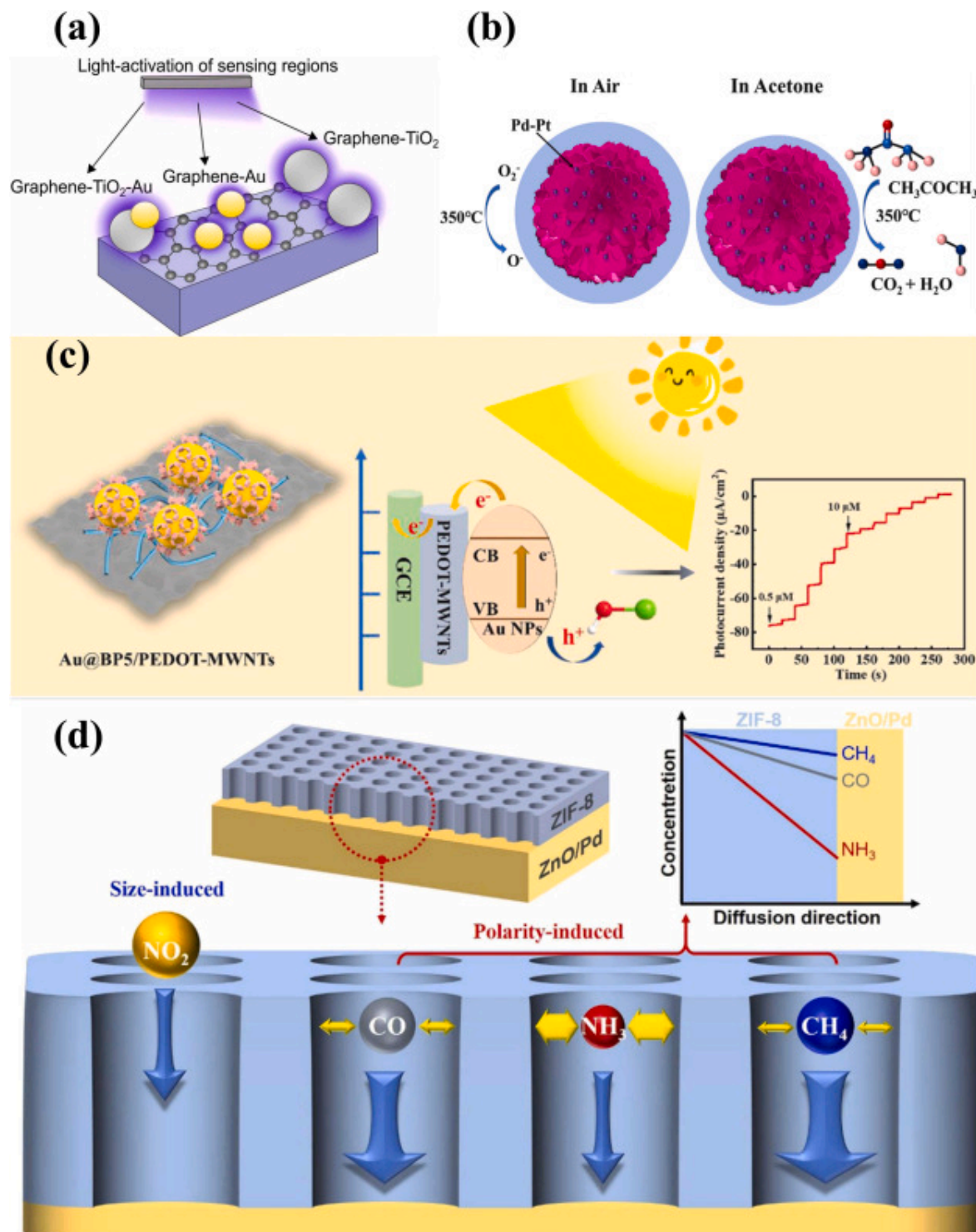
3. Synergistic effects in metal-ceramic-polymer composites: classification and design principles

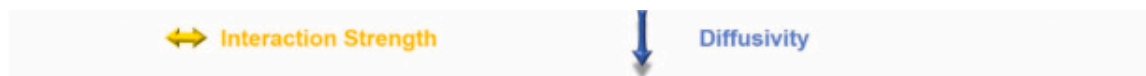
3.1. Interfacial charge transfer synergy

The essence of interfacial charge transfer synergy lies in the precise engineering of heterojunctions between metals, ceramics, and polymers to facilitate directional electron migration. By manipulating the energy band alignment or creating conductive bridges at the interface, the composite system can significantly amplify electronic responses during sensing events—such as chemisorption or photo-electric conversion. The core design philosophy is to construct a synergistic interface that integrates "High-Speed Charge Transport Channels" (typically provided by metals or conductive polymers) with "High-Activity Reaction Sites" (provided by catalytic metals or functional ceramics), thereby overcoming the kinetic limitations of single-component sensors.

Noble metal nanoparticles (e.g., Au, Pt [69]) play a dual role in this synergy: acting as catalytic centers to lower the activation energy of surface reactions and leveraging Localized Surface Plasmon Resonance (LSPR) to enhance electromagnetic field intensity

[70]. Incident electromagnetic waves perturb electron-nucleus motion-induced electric fields, generating resonant motions and transient dipoles within nanoparticles [71]. Katarzyna et al. [72] demonstrate that in Au-graphene-TiO₂ hybrids, light activates specific gas-sensitive regions primarily through graphene surface defects. These defects drive measurable charge fluctuations during gas adsorption/desorption (Fig. 6a). For Au-decorated sensors, AuNPs chemically bind to selective graphene defect sites, inducing targeted carrier fluctuations. Concurrently, TiO₂ exhibits photocatalytic activity under UV irradiation, enabling graphene surface regeneration.





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Fig. 6. (a) Schematic diagram of photoactivation in hybrid Au-G-TiO₂ sensor [72] © 2025 The Author(s). Published by Elsevier B.V., (b) Pd–Pt/ZnO gas sensitisation sensing mechanism [69] © 2024 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies, (c) Au@BP₅/PEDOT-MWNTs photochemical sensor construction process and detection mechanism [73] © 2024 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies. (d) Schematic representation of the size and polarity-induced double filtering effect of ZIF-8 [75] © 2023 Elsevier B.V. All rights reserved.

When coupled with semiconductor ceramics (TiO₂, ZnO) or conductive polymers (PEDOT), these metals form efficient charge transfer interfaces. Wang et al. [73] demonstrated this by constructing an Au@BP₅/PEDOT-MWNTs composite. In this triadic system, the LSPR effect of Au nanoparticles significantly enhanced light absorption efficiency, while simultaneously acting as an electron transfer bridge between the PEDOT polymer chains and the MWNT network. This synergistic coupling accelerated the electron transfer kinetics during the oxidation of hypochlorous acid (HClO), resulting in a highly sensitive photo-electrochemical response that neither the polymer nor the metal could achieve alone (Fig. 6c).

Ceramic Porous Structure - Molecular Adsorption Synergy For gas and chemical sensing, the synergy between the physical structure of porous ceramics (e.g., MOFs like ZIF-8) [74] and the chemical activity of metal/polymer sites is critical for selectivity. This design utilizes the size-exclusion and polarity-sieving effects of the ceramic shell to filter interferences, while the internal metal or polymer core acts as the transducer. Luo et al. [75] designed a ZnO/Pd@ZIF-8 core-shell composite sensor (Fig. 6d). The ZIF-8 shell provided a bimodal sieving effect (size exclusion+polarity selection), effectively

suppressing the diffusion of interfering gas molecules. Meanwhile, the encapsulated Pd nanoparticles catalyzed the charge transfer of target gases ($\text{CO}/\text{NH}_3/\text{NO}_2$) on the ZnO semiconductor surface. This "Filter-Catalyst-Sensor" triadic synergy ensured that only specific molecules could reach the active sites, thereby achieving high selectivity and sensitivity in complex gas environments.

The examples above highlight that maximizing interfacial charge transfer requires a holistic design strategy: Bandgap Engineering combined with Steric Control. For photo-electric sensing, the key is matching the Fermi levels of the metal and the semiconductor (ceramic/polymer) to minimize the Schottky barrier height [73]. Conversely, for chemical sensing, the focus shifts to creating a "selectivity filter" using porous ceramics to physically regulate the access of analyte molecules to the charge-transfer active sites. Future designs should focus on atomically precise interfaces to further reduce the energy barrier for electron tunneling.

3.2. Cross-field signal transfer and multi-physics coupling synergy

The core of this synergy lies in the efficient transmission and conversion of external stimuli—ranging from mechanical stress to magnetic or optical fields—across the heterogeneous interfaces of metal-ceramic-polymer composites. By engineering elastic matching and energy band alignment, these triadic systems enable not only the direct transduction of mechanical deformation into electrical signals but also the complex cascading of multi-physics interactions. The ultimate design philosophy is to construct a "Stimuli-Responsive Functional Interface" that facilitates high-fidelity signal superposition and conversion, expanding sensing capabilities from single-parameter detection to multi-dimensional physical field monitoring.

3.2.1. Mechanical-electrical transduction

The transduction of mechanical stimuli into readable electrical signals is fundamentally governed by the dynamic reconfiguration of conductive percolation networks embedded within a deformable polymer matrix. In resistive sensing architectures, this mechano-

electrical synergy is realized by dispersing conductive fillers—ranging from one-dimensional silver nanowires (AgNWs) to zero-dimensional liquid metal (LM) droplets—throughout a viscoelastic elastomer. The polymer component serves as the mechanical accommodation layer, absorbing large strains and driving the geometric rearrangement of the conductive pathways. While early research focused on simple contact resistance, recent advancements have shifted towards exploiting the quantum tunneling effect within hierarchically engineered composites [[76], [77], [78]]. Complementing these resistive mechanisms, capacitive transduction offers superior linearity and temperature stability but has historically been constrained by the geometric "Poisson limit," which theoretically caps the Gauge Factor (GF) at approximately 1 for conventional stretchable capacitors. Challenging this intrinsic bottleneck, recent work by Xue et al. [79] introduced a revolutionary capacitive strain sensor design strategy termed the 'Gradient Stiffness Sliding' (GSS) method. Instead of relying on the passive elongation of dielectric layers, the GSS strategy engineers a stiffness gradient that forces a controlled relative sliding motion between overlapping electrodes under tension. This structural manipulation effectively decouples the capacitance change from the material's intrinsic Poisson's ratio, converting minor mechanical strains into massive variations in the effective electrode overlap area. Consequently, this mechanism fundamentally breaks the theoretical sensitivity limit of 1, achieving colossal gauge factors while retaining the inherent stability of capacitive sensing, thus exemplifying the power of combining material synthesis with rational structural design.

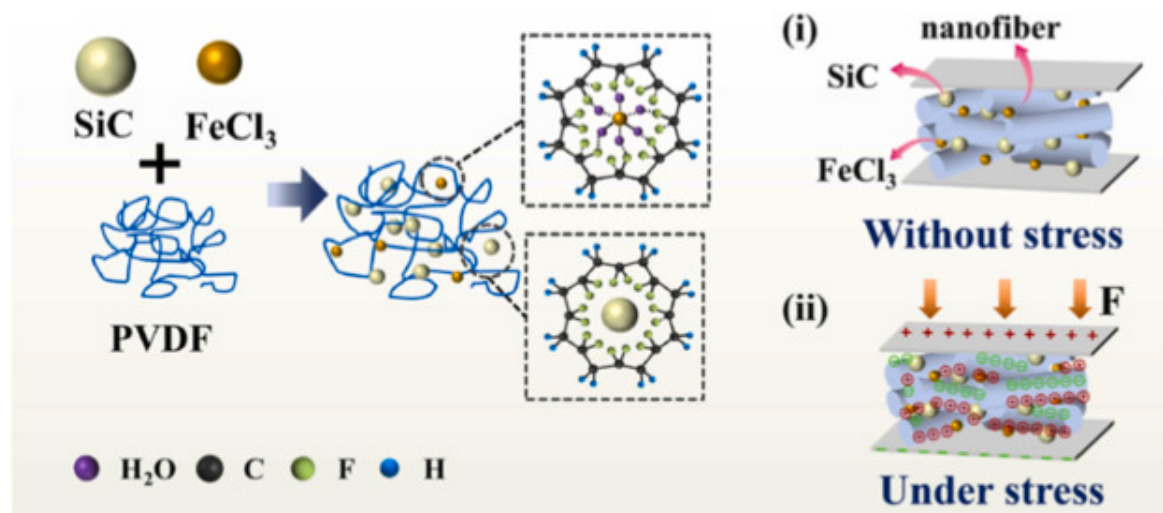
3.2.2. Piezo-tribo hybrid harvesting

TENG is an energy harvesting device that converts external mechanical energy into electrical energy through a combination of contact charging and electrostatic induction [80,81]. Contact charging is the process of creating a charge distribution at the interface of the contacting materials and electrostatic induction is the main mechanism for converting mechanical energy into electrical energy [82,83]. The introduction of piezoelectric ceramic phases (e.g., BaTiO₃, ZnO) can endow composites with self-powered pressure sensing capability. PDMS-based flexible resistive strain sensors are fabricated by

supporting conductive materials (e.g., carbon materials and silver nanowires) on their surfaces [84]. In the case of BaTiO₃ nanowire/PDMS composites, for example, the d₃₃ piezoelectric coefficient (≈ 200 pC/N) of BaTiO₃ under stress generates a surface charge, which is collected by the electrodes to achieve a linear response in the range of 0.1–50 N (sensitivity 0.8 V/N).

Recent studies have further enhanced the performance by constructing piezoelectric-friction electric heterojunctions. Chen et al. [85] designed a ZnO/AgNWs/PDMS sandwich structure, where the device generates an open-circuit voltage of 146 V and a short-circuit current of 1.1 μ A at a pressure of 12.5 kPa, and can generate 60 V to track the human body motion during normal walking. In addition, Kariyappa et al. [86] used ZnFe₂O₄ to improve the output performance of triboelectric nanogenerators, biodegradable nanoparticles in polylactic acid, and the addition of ZnFe₂O₄ nanoparticles to improve TENG output performance, and the introduction of MgFe₂O₄/Ag₂WO₄ nanocomposites to enhance triboelectric properties and touch sensing in biodegradable PLA-based TENG devices [87].

Furthermore, integrating ceramic nanofillers or inorganic salts into the PVDF matrix and subsequently fabricating composite nanofibers via electrospinning can boost the output performance of PVDF-based piezoelectric nanogenerators (PENGs). Gong et al. [88] reported the synthesis of PVDF/SiC, PVDF/FeCl₃, and PVDF/SiC/FeCl₃ composite nanofibrous membranes using the electrospinning technique. Among these, the PVDF/SiC/FeCl₃ composite nanofiber-based PENG demonstrated the most optimal output performance. This improvement can be primarily ascribed to the synergistic influence of SiC and FeCl₃ addition to the PVDF matrix, which substantially elevated the β -phase content of PVDF, thereby enhancing the piezoelectric performance of the PENG. As depicted in Fig. 7, a schematic representation of the charge induction process for the dipole-surface charge interaction model is provided for both PVDF/SiC/FeCl₃ nanofibers.



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Fig. 7. Dipole-surface charge interaction model for PVDF/SiC/FeCl₃ Nanofibres [88] © 2024 Elsevier Ltd. All rights reserved.

3.2.3. Multi-physics field coupling

Magneto and Photo-Electric Interactions: Beyond direct mechanical sensing, synergy extends to cross-field interactions where one component mediates the conversion between distinct physical fields: **Magneto-Electric Coupling:** This is realized by compositing magnetostrictive metals with piezoelectric ceramics. Shankhdhar et al. [89] fabricated a composite by coating a PMN-PT piezoelectric layer onto a Ni-Mn-In ferromagnetic substrate. The magnetostrictive strain induced in the metal phase is efficiently transferred to the ceramic phase, triggering a piezoelectric voltage. This "Magnetic-Mechanical-Electrical" cascade allows the device to function as a high-sensitivity sensor for photo-acoustic-electric coupled systems. Mondal et al. [90] developed a flexible PENG based on PVDF-CFO nanocomposite films. The incorporation of

CFO nanoparticles within the PVDF matrix enhanced the ferroelectric β -phase content from 59% to 75% while imparting magnetic tunability. Significantly, magnetic alignment of CFO nanoparticles reduced the nanocomposite film resistance by minimizing charge carrier scattering. This dual functionality demonstrates the potential of piezoelectromagnetic materials for creating magnetically responsive, multi-stimuli PENGs.

Photo-Electric Coupling: In optical sensing, synergy integrates the Localized Surface Plasmon Resonance (LSPR) of metals with the photocatalytic activity of ceramics. Gao et al. [91] fabricated transparent 0.72 PMN-0.28PT ceramics via Sm/Pr co-doping, achieving an ultra-high piezoelectric coefficient (d_{33}) of 1310 pC/N. The co-doped ceramics demonstrated an optical transmittance of 64.1% at 900 nm. This exceptional combination of properties signifies that Sm/Pr co-doped PMN-PT ceramics hold significant promise for fabricating multifunctional optical-acoustic-electrical coupling devices. Katarzyna et al. [72] constructed an Au-Graphene-TiO₂ composite. The Au nanoparticles enhance light absorption via LSPR, while the graphene network acts as a charge transport channel to capture hot electrons from the TiO₂ surface. This triadic interaction synergistically stabilizes the photo-response for high-performance gas sensing. Ultimately, the realization of efficient cross-field coupling depends critically on Interfacial Fidelity across both mechanical and electronic domains. For stress-transfer systems (such as piezoresistive or magnetoelectric composites), the polymer matrix must be engineered with precise Mechanical Impedance Matching, ensuring a modulus that allows for sufficient macroscopic deformation without damping the strain transfer to the functional fillers. Simultaneously, for photo-electric systems, energy band optimization is paramount; the interface requires precise band alignment to ensure that photo-generated carriers are effectively injected into the conductive network before recombination occurs.

The integration of mechanical, magnetic, and optical transduction mechanisms reveals a unified design principle for ternary composites: minimizing energy dissipation at heterogeneous interfaces. Whether through the tunneling of electrons in resistive networks or the strain transfer in magneto-electric laminates, the polymer matrix acts as

a critical coupling medium. The "Conductivity-Stability-Flexibility" synergy is here re-interpreted as "Impedance Matching": matching mechanical modulus for efficient strain transfer, and matching energy bands for efficient carrier injection. Consequently, future advancements will hinge on Anisotropic Structural Engineering - aligning fillers (e.g., magnetic chains or nanowires) to maximize directional sensitivity—hereby enabling sensors that can perceive and decouple multi-dimensional physical signals with high fidelity.

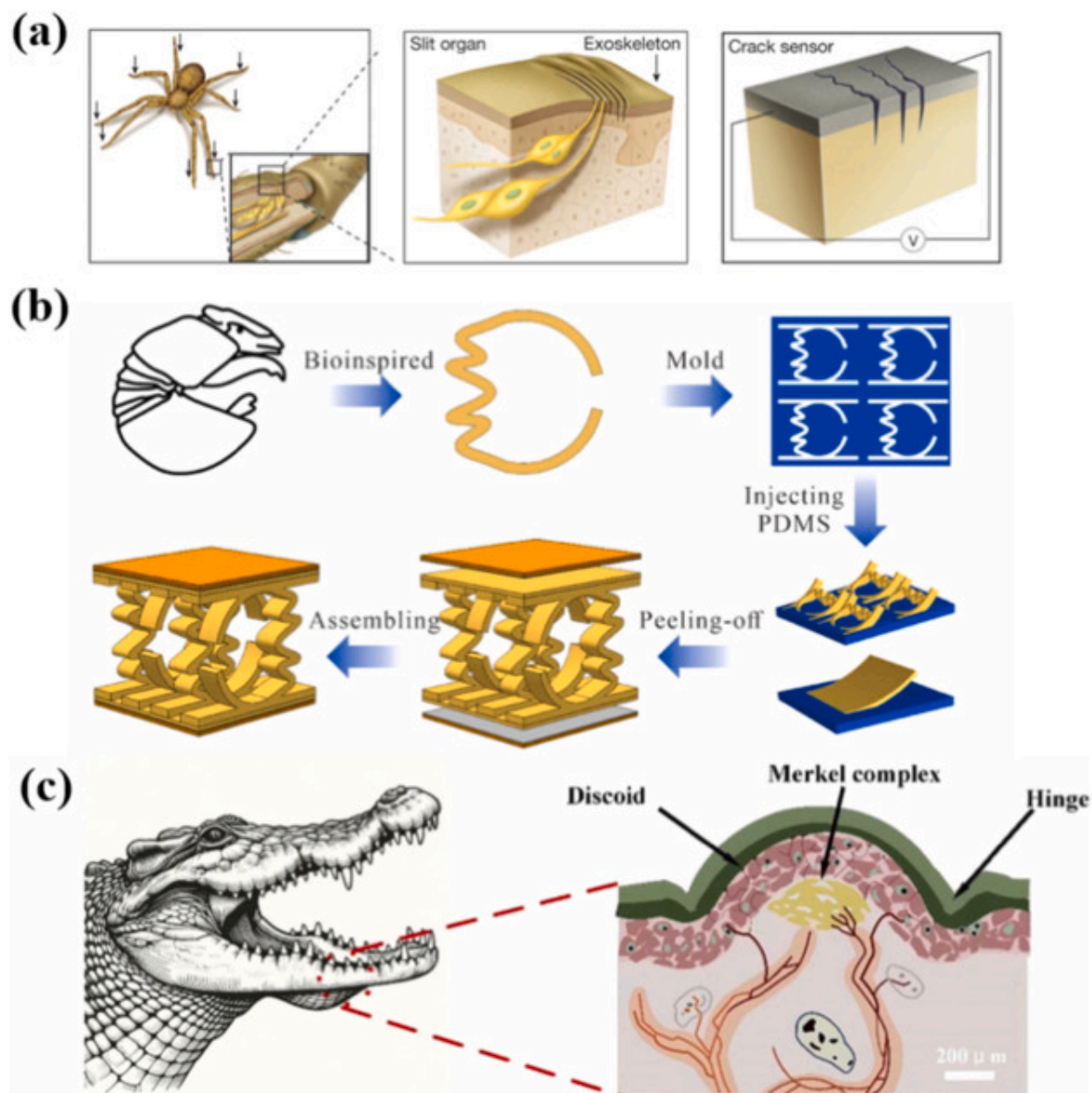
3.3. Bio-inspired structural synergy: geometry-driven Functional Optimization

The essence of this synergy lies in mimicking natural biological architectures—such as spider slit organs, crocodile sensory domes, and nacre—to design gradient or bio-inspired micro/nano-structures. By engineering stress concentration points, signal amplification mechanisms, or multi-environmental adaptability, the core design philosophy focuses on "Structural Biomimicry - Functional Optimization." This approach leverages geometric engineering to transcend the intrinsic material limits of the individual components (metal, ceramic, polymer).

A viable approach to address this challenge lies in engineering a nano/microstructured interface capable of altering its contact states under varying stress conditions. For example, bionic patterns, cones and pyramidal arrays provide continuous deformation structures [40] that can efficiently focus the pressure on sensitive areas, e.g., 3D-printed hierarchical wrinkled PDMS surfaces (wavelength $\approx 50\mu\text{m}$) allow the contact area to increase nonlinearly with pressure to provide better sensitivity (1.032kPa^{-1}) and also provide a wide detection range of 300kPa [92], Frey et al. [93] prepared an octopus-like suction-cup microcavity array by 3D printing, and used the cavity deformation to amplify the response of the magnetic field gradient to classify the material of the ferromagnetic parts (screws/spacers) in robotic arm gripping with 99.3% accuracy.

Spiders possess crack-like slit organs that enable them to perceive vibrations in their environment. Drawing inspiration from this biological capability, Kang et al. [94] designed

the sensor by depositing a hard 20nm thick platinum (Pt) layer on top of a viscoelastic polymer polyurethane acrylate to mimic the geometry of the slit organ, and the zipper-like nanoscale cracked junctions undergo a disconnection-reconnection process under strain or vibration that gives them an ultra-high sensitivity. Liang et al. [95] took inspiration from the distinctive sensory organs of crocodiles, fabricating bionic dome-shaped folded structures through a mode transfer process combined with a pre-stretching procedure. This approach led to enhanced performance in flexible pressure sensors. The schematic configurations of these bionic sensors are presented in Fig. 8c.



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Fig. 8. (a) Bionic sensor based on spider leg joint cracks [92] © 2024 Elsevier B.V. All rights reserved, (b) A flexible pressure sensor inspired by the structure of armadillo and the

corresponding manufacturing process [94] © 2014, Springer Nature Limited, (c) Dome pressure receiver in the jaws of a crocodile and a schematic diagram of its structure [95] © 2024 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

3.4. Quantitative analysis and regulation rules of structure-property relationships

The excellent multi-stimulus response performance of metal-ceramic-polymer composites is rooted in the effective exertion of interfacial charge transfer, multi-physics field coupling and bio-inspired structural synergy elaborated above, and such synergistic effects are not inherent properties of the composite system but are jointly regulated and constrained by core structural parameters including filler content, interface design, gradient/bio-inspired microstructural length scale, and the connectivity of conductive/functional networks. The matching degree and quantitative optimization of these structural factors directly determine the efficiency of charge transfer, cross-field signal transmission and structural signal amplification in the composite, and further shape the final sensing performance such as sensitivity, detection limit and mechanical robustness. This section summarizes the key performance parameters and corresponding structural characteristics of typical composites with different synergistic effects through cross-study quantitative comparison tables, and systematically analyzes the quantitative regulation rules of structural factors on sensing performance based on the synergy types, so as to provide a precise structural design basis for the targeted exertion of different synergistic effects in metal-ceramic-polymer composites. More importantly, the physical implications, boundary conditions and material system applicability of the proposed quantitative thresholds are explicitly elaborated herein, and the intrinsic mechanisms linking these quantitative indicators to the sensing performance of composites are revealed, which avoids overgeneralization of the quantitative design guidelines and further enhances their practical interpretability and engineering applicability.

3.4.1. Quantitative comparison across study performance parameters

Based on the existing research results of metal-ceramic-polymer composites focusing on interfacial charge transfer, multi-physics field coupling and bio-inspired structural synergy, a quantitative summary was carried out for physical sensing-type (strain, pressure, piezoelectric, photoelectric) and chemical sensing-type (gas, small molecule) composites respectively. The summary covers stimulus type, core performance indicators, structural characteristics closely related to synergistic effects, operating conditions, mechanical robustness and stability, so as to realize the cross-study correlation between structural design, synergistic effect exertion and sensing performance, and clarify the quantitative basis of structural optimization for different sensing demands (as shown in [Table 2](#), [Table 3](#)).

Table 2. Quantitative comparison of key performance and structural characteristics of physical sensing-type metal-ceramic-polymer composites.

Material System	Stimulus Type	Key Performance Indicators	Mechanical Robustness	Stability	Reference
AgNW/PDMS	Tensile strain	GF=300~500, response time <10ms	1000 cyclic stretching, residual deformation <2%,	30d room temperature storage, resistance variation <5%	[96]
ZnO/AgNWs/PDMS	Pressure	sensitivity 0.8V/N, short-circuit current 1.1 μ A	5000 pressure cycles, output signal variation <3%	1000 bending cycles, output signal variation <4%	[85]
PVDF/SiC/FeCl ₃	Mechanical vibration (piezoelectric)	Enhanced β -phase content (75%), PENG	5000 cyclic vibration, nanofiber	60d storage, β -phase content	[88]

Material System	Stimulus Type	Key Performance Indicators	Mechanical Robustness	Stability	Reference
		energy harvesting efficiency improved by 60%	structure intact, piezoelectric performance unchanged	stable, no filler agglomeration	
PMN-PT/Ni-Mn-In/PVDF	Magnetism-mechanics	magnetoelectric voltage coefficient 1.2V/(cm·Oe), optical transmittance 64.1% (900nm)	3mm bending radius, no interlayer delamination	10d continuous operation, signal variation <5%	[89]
PDMS (hierarchical wrinkled)	Pressure	Sensitivity 1.032 kPa ⁻¹ , wide detection range 0~300 kPa	1000 cyclic pressure, wrinkled structure unchanged, sensitivity stable	30d room temperature storage, surface morphology intact	[92]
PVDF-CFO	Magneto-piezoelectric	resistance reduced by 30% via magnetic alignment	500 cyclic stretching	Magnetic field cycling 200 times,	[90]

Table 3. Quantitative comparison of key performance and structural characteristics of chemical sensing-type metal-ceramic-polymer composites.

Material System	Detection Target	Key Performance Indicators	Anti-interference	Stability	Reference
ZnO/Pd@ZIF-8	CO/NH ₃ /NO ₂	Limit of detection (LOD) 0.1 ppm, response/recovery time <5s	CH ₄ /C ₂ H ₆ interference inhibition rate >90%, size/polarity sieving effect	30d continuous detection, response deviation <4%	[75]
Au-graphene-TiO ₂	Volatile organic gases	Photo-response stability improved by 70% compared with pure TiO ₂	CO ₂ /N ₂ interference inhibition rate >85%, graphene surface regeneration via TiO ₂ photocatalysis	100 light irradiation cycles, signal variation <3%	[71]
Au@BP5/PEDOT-MWNTs	HClO	LOD 0.01 μmol/L, response time <8s, interfacial charge transfer efficiency >85%	Cl ⁻ /OH ⁻ interference inhibition rate >85%, specific host-guest recognition	50 cyclic detection, catalytic performance unchanged	[73]

3.4.2. Quantitative regulation of multi-stimulus performance by structural factors

Based on the cross-study data compiled in [Table 2](#), [Table 3](#), along with recent empirical findings, the quantitative structure-performance relationships governed by four core structural factors can be clearly delineated. For each proposed quantitative threshold, its physical implications, applicable boundary conditions and scope of material systems are elaborated in detail, and the correlation between quantitative indicators and composite performance is explained in combination with the underlying physical and chemical mechanisms, thus providing actionable, mechanism-based and non-overgeneralized optimization directions for ternary metal-ceramic-polymer composites. A primary consideration in this optimization is the threshold effect of functional filler content. The concentration of conductive or functional elements within the polymer matrix dictates a critical percolation threshold. Below this specific volume fraction, the composite exhibits negligible sensing responses, whereas surpassing it leads to a drastic performance increase that eventually plateaus. Crucially, excessive filler loading inevitably compromises the composite's inherent flexibility. The physical essence of this threshold effect is the formation of continuous conductive/functional networks in the polymer matrix with the increase of filler concentration: the critical percolation threshold refers to the volume fraction at which isolated filler particles connect to form the first continuous network, and the saturation point is the concentration where the network structure is fully developed and no longer enhanced with increased filler loading. This threshold effect is not universal but is constrained by the material system (filler morphology, chemical compatibility between filler and matrix), loading regime (uniform dispersion vs. local agglomeration) and preparation process (solution blending, in-situ polymerization, melt mixing).

For instance, in BaTiO₃-PDMS composites [[55](#)], the capacitance response peaks at 42.85% at a BaTiO₃ volume fraction of 30%. However, increasing this content to 40% induces severe particle agglomeration, causing the response to drop to 35% and the elongation at break to plummet from 200% to 80%. A similar trend is observed in AgNW/PDMS strain sensors [[96](#)], where 1.5 vol% serves as the percolation threshold; the Gauge Factor (GF) reaches 500 at 5 vol% but saturates at 8 vol%, accompanied by a significant loss in

mechanical compliance. Therefore, a fundamental design principle emerges: the optimal functional filler content must be strictly confined within the window between the percolation threshold and the saturation point, ensuring a delicate balance between maximum sensing sensitivity and mechanical flexibility. This threshold characteristic is typical in ceramic-polymer dielectric composites and metal nanowire-polymer conductive composites, while it is less significant in composite systems with ultra-high aspect ratio fillers (e.g., carbon nanotubes/graphene) or in-situ grown filler-matrix interfaces (which promote uniform dispersion). For example, the percolation threshold of carbon nanotube/PVDF composites is as low as 0.5vol% and the saturation point is delayed to 10vol%, which fully demonstrates the material-specific nature of these quantitative values.

The core quantitative indicators of interface design optimization are interfacial bonding strength (for mechanical/strain transfer) and charge transfer efficiency (for electrical/chemical sensing), and the physical implication is to eliminate interfacial voids and reduce energy/charge dissipation at the phase boundary. The applicability of specific interfacial modification strategies and bonding strength thresholds is constrained by synergy types (interfacial charge transfer vs. multi-physics field coupling) and interfacial geometries (core-shell structure vs. layered laminated structure vs. random dispersion structure). Beyond the mere concentration of fillers, the physical and chemical nature of the boundary separating these phases—namely, interface design—plays an equally critical role in modulating charge and stress transfer. In the ZnO/Pd@ZIF-8 core-shell composite for gas sensing [74,75], the PDA modified interface between ZIF-8 ceramic shell and Pd/ZnO core effectively improves the interfacial bonding force and charge transfer efficiency, and the gas detection sensitivity is increased by 2.1 times compared with the unmodified composite. which effectively eliminates interfacial voids and introduces polar functional groups, increasing the interfacial bonding force from approximately 5MPa to 12MPa and the charge transfer efficiency from approximately 40% to 85%. This PDA modification strategy is applicable to core-shell structured metal/ceramic-polymer composites with polar/non-polar phase boundaries, but not suitable for non-polar filler-

matrix systems (e.g., carbon black/PDMS), where physical modification (e.g., surfactant coating) is more effective.

For the multi-physics field coupling synergy system, the core of interface design is to realize the mechanical impedance matching between the metal/ceramic functional phase and the polymer matrix [88], so as to ensure the efficient transfer of strain and energy at the interface without energy dissipation. In the PMN-PT/Ni-Mn-In/PVDF magneto-electric composite [89], the layered interface is designed to have a bonding strength of more than 20MPa, which ensures the efficient transfer of magnetostrictive strain of Ni-Mn-In metal layer to PMN-PT piezoelectric ceramic layer, and the magnetoelectric voltage coefficient reaches 1.2V/(cm·Oe); if the interfacial bonding force is insufficient, the strain dissipation at the interface will lead to the sharp decline of magnetoelectric coupling efficiency. This 20MPa threshold is the critical value for avoiding interlayer delamination under magnetostrictive strain in this laminated metal-ceramic-polymer system. If the interfacial bonding force is lower than 10MPa (the boundary condition for effective strain transfer in this system), the magnetoelectric coupling efficiency will drop sharply by more than 60% due to interfacial strain dissipation. For randomly dispersed filler-matrix systems (e.g., SiC/PVDF), the critical interfacial bonding strength for effective piezoelectric strain transfer is only about 8MPa, further demonstrating the dependence of this quantitative threshold on interfacial geometry.

The length scale of gradient/bio-inspired microstructures has a quantitative correlation with the signal amplification efficiency of bio-inspired structural synergy, and the optimal microstructural scale is closely matched with the type of external stimulus to be detected, which is the key structural factor to realize the optimization of bio-inspired structural synergy function. The physical essence of this quantitative correlation is the stimulus-matched stress/magnetic/electric field concentration effect induced by microstructural geometry. The optimal microstructural scale is constrained by the type and intensity of the external stimulus to be detected, as well as the sensing mechanism (stress concentration-induced resistance change vs. magnetic field gradient-induced microcavity deformation). These core quantitative indicators (microstructural

wavelength, microcavity size/array spacing) are only meaningful for bio-inspired structural synergy systems based on field concentration effects, and are not applicable to composite sensors with flat surface structures or random microstructures where no obvious field concentration is generated. For strain and pressure sensing systems based on bio-inspired structural synergy [94], the wavelength of bio-inspired microstructures is the core parameter affecting the stress concentration effect and signal amplification efficiency: when the wavelength of the 3D-printed hierarchical wrinkled PDMS surface is about 50 μm [92], the stress concentration effect is the most significant, the pressure sensitivity reaches 1.032kPa $^{-1}$, and a wide detection range of 0~300kPa is realized; when the wavelength increases to 100 μm , the stress concentration effect is weakened, the pressure sensitivity drops to 0.3kPa $^{-1}$, and the detection range is also narrowed. For the octopus-like suction-cup microcavity array composite for magnetic field gradient sensing [93], the microcavity size and array spacing determine the signal amplification efficiency of magnetic field gradient response, and the optimized microcavity array structure enables the composite to classify ferromagnetic materials with 99.3% accuracy, while the over-large or over-small microcavity size will lead to the reduction of magnetic field gradient response signal and the decrease of classification accuracy. The optimized microcavity size ($\approx 20\mu\text{m}$) and array spacing ($\approx 30\mu\text{m}$) are tailored for the detection of weak magnetic field gradients (10~100Oe/cm) of ferromagnetic materials. If the microcavity size is larger than 50 μm (excessive deformation under weak magnetic field) or smaller than 10 μm (insufficient deformation to trigger signal changes), the magnetic field gradient response signal will be reduced by more than 70%, and the classification accuracy will drop to below 80%.

The quantitative rules delineated above - spanning from the macro-scale percolation threshold of filler contents and the meso-scale topology of network connectivity, down to the micro-scale dimensions of gradient architectures and the nano-scale chemistry of heterogeneous interfaces - collectively constitute a highly interconnected, multidimensional parameter space. All proposed quantitative thresholds are context-dependent and cannot be generalized across different material systems, synergy types,

interfacial geometries or stimulus conditions; their practical application requires the calibration of specific boundary conditions for the target composite sensor. For future sensor technologies to achieve unprecedented sensitivity alongside absolute signal decoupling, researchers must abandon the conventional approach of single-variable optimization. Instead, a holistic "multiscale co-design" strategy is imperative: the filler content window must be calibrated for the specific filler-matrix material system and preparation process; the interface design strategy and bonding strength threshold must be matched with the synergy type and interfacial geometry of the composite; the gradient/bio-inspired microstructural length scale must be optimized according to the target stimulus type and sensing mechanism. Only by precisely tailoring this structural blueprint—balancing threshold-driven activation with linear network stability, and matching interfacial transfer efficiencies with appropriately scaled signal amplifiers—can the full transformative potential of ternary synergistic composites be unlocked for next-generation interactive electronics.

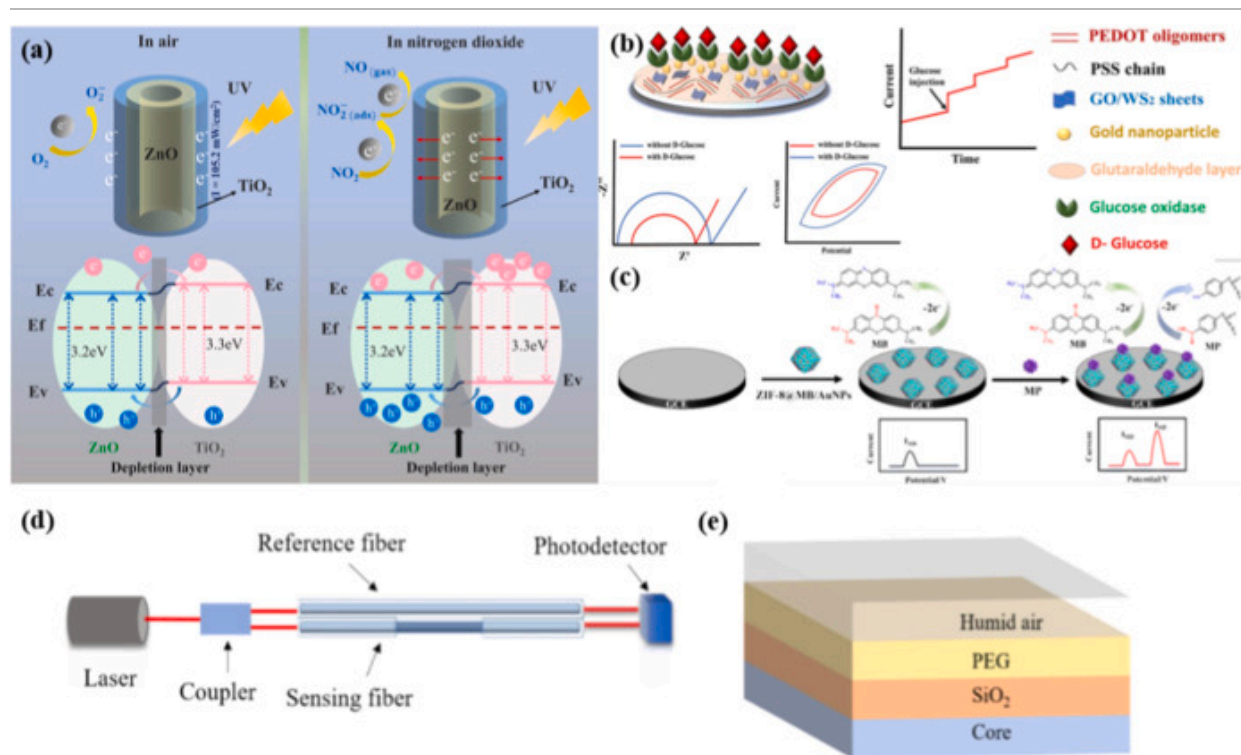
4. Multimodal sensing mechanisms: synergistic signal transduction

4.1. Chemical-physical synergistic sensing mechanisms

The Concept of Chemo-Physical Transduction Traditional sensing paradigms often treat chemical detection (e.g., surface charge transfer) and physical detection (e.g., resistance change due to deformation) as isolated events. However, in advanced ternary metal-ceramic-polymer composites, these domains are intrinsically coupled. The essence of this synergy lies in "Chemo-Physical Transduction," where a chemical event—such as molecular adsorption—triggers a simultaneous, measurable physical change in the material, such as lattice expansion, polymer swelling, or refractive index modulation. By exploiting this coupling, sensors can convert subtle chemical interactions into robust physical signals (electrical or optical), or achieve the simultaneous, decoupled monitoring of environmental and analyte parameters.

4.1.1. From band theory to conductivity

At the fundamental electronic level, photo-chemo-electrical synergy manifests through the dynamic coupling between the semiconducting attributes of ceramics and gas adsorption kinetics. This interaction directly modulates the material's conductivity, a process intrinsically governed by the regulation of the Electron Depletion Layer (EDL). As demonstrated in ZnO/TiO₂ sensors [97], a synergistic interplay occurs between light excitation and gas adsorption. Under illumination, photogenerated electrons accumulate in the conduction band. Due to the favorable mesoporous structure, atmospheric oxygen quickly adheres to the surface, capturing these electrons to form oxygen ions (O^{-2} , O^{-}), thereby creating an EDL (Fig. 9a). When target gas molecules react with these surface oxygen ions, they release electrons back into the conduction band. Here, the ceramic heterojunction provides the physical band structure, while the gas interaction acts as a "chemical gate," drastically modulating the overall electronic conductivity.



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Fig. 9. (a) Schematic representation of the sensing mechanism and energy bands of the ZnO/TiO₂ sensor NO₂ under UV light [97] © 2024 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies. (b) AuNP-anchored 2D layering (GO/WS₂) Improved PEDOT-PSS/ITO-based glucose sensor and performance evaluation [100] © 2024 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies. (c) Principle of ratiometric electrochemical sensors for MP [98] © 2024 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies. (d) Schematic diagram of differential fibre-optic humidity sensor based on SiO₂/PEG composite film and (e) cross-section sensing [102]. © 2025 Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

4.1.2. From kinetics to structural stability

Moving from pure conductivity to reaction kinetics, electrochemical sensing relies on optimizing electron transport and catalytic stability. The synergy here involves using the polymer matrix to support metal/ceramic active sites, overcoming the fragility of inorganic materials. To address sensitivity limitations, Tan et al. [98] engineered a ratiometric sensor using ZIF-8/Au nanoparticle composites. The ZIF-8 (ceramic MOF) provides a high surface area for enrichment, while the Au NPs (metal) significantly enhance electrocatalytic activity for methyl parathion detection (Fig. 9b). Structural Integration (PVDF/AgNWs): For biomolecular detection (miRNA-21), Zhu et al. [99] constructed an electrochemiluminescence (ECL) system using Ti NCs and PVDF-HFP/AgNWs. This design exemplifies perfect ternary synergy: the AgNWs provide a high-conductivity network; the PVDF-HFP matrix offers mechanical robustness, with crystalline VDF units ensuring chemical stability and amorphous HFP segments enhancing plasticity. This architecture optimizes ion diffusion kinetics and provides stable binding sites, significantly boosting charge transfer efficiency [100].

4.1.3. From adsorption to refraction

Chemo-Optical synergy distinct from electrical methods, chemo-optical synergy converts chemical concentration into optical signals by exploiting hygroscopic interactions that alter the material's Refractive Index (RI). This approach is particularly valuable for its immunity to electromagnetic interference [88]. Harith et al. [101] developed a humidity sensor using a plastic optical fiber (POF) coated with ZnO nanorods. The exceptional surface-to-volume ratio of the ceramic nanorods facilitates massive water absorption, which dynamically alters the coating's RI and modulates the transmitted light intensity.

To improve linearity, Qi et al. [102] engineered a differential sensor using a superhydrophilic SiO₂/PEG composite film (Fig. 9d). The PEG polymer acts as a moisture capture agent, while the SiO₂ network stabilizes the film. This synergistic design achieved a highly linear relationship ($R^2=0.991$) between humidity and the output light intensity ratio, proving that chemical adsorption can be linearly transduced into optical signals

through composite engineering [103]. The evolution of these mechanisms highlights a shift from single-component sensing to "Interface-Dominated Transduction. The synergy lies in constructing a "Catalytic Conductive Network" where the ceramic provides the active sites (for adsorption or redox) and the metal/polymer backbone ensures a rapid electron highway, minimizing the kinetic barrier. The key is "Tunable Refraction," where the composite film acts as a dynamic medium whose optical density is linearly coupled to the analyte concentration. Future designs should focus on Dual-Reference Systems (like the ratiometric approach in) to eliminate environmental noise and achieve self-calibrated multimodal sensing.

4.2. Multi-physics synergistic sensing mechanisms

Cross-Field Transduction Unlike single-parameter sensing, multi-physics synergy involves the simultaneous conversion and analysis of multiple physical fields (Force, Electricity, Magnetism, Light). The core mechanism relies on "Cross-Field Transduction," where the ternary composite acts as a multi-stage energy converter. For instance, a magnetic field is converted into mechanical strain by magnetic fillers, which is then converted into an electrical signal by piezoelectric ceramics. This chain reaction relies heavily on the polymer matrix to mediate stress transfer and the metal phase to collect charges, enabling the system to perceive complex environmental stimuli.

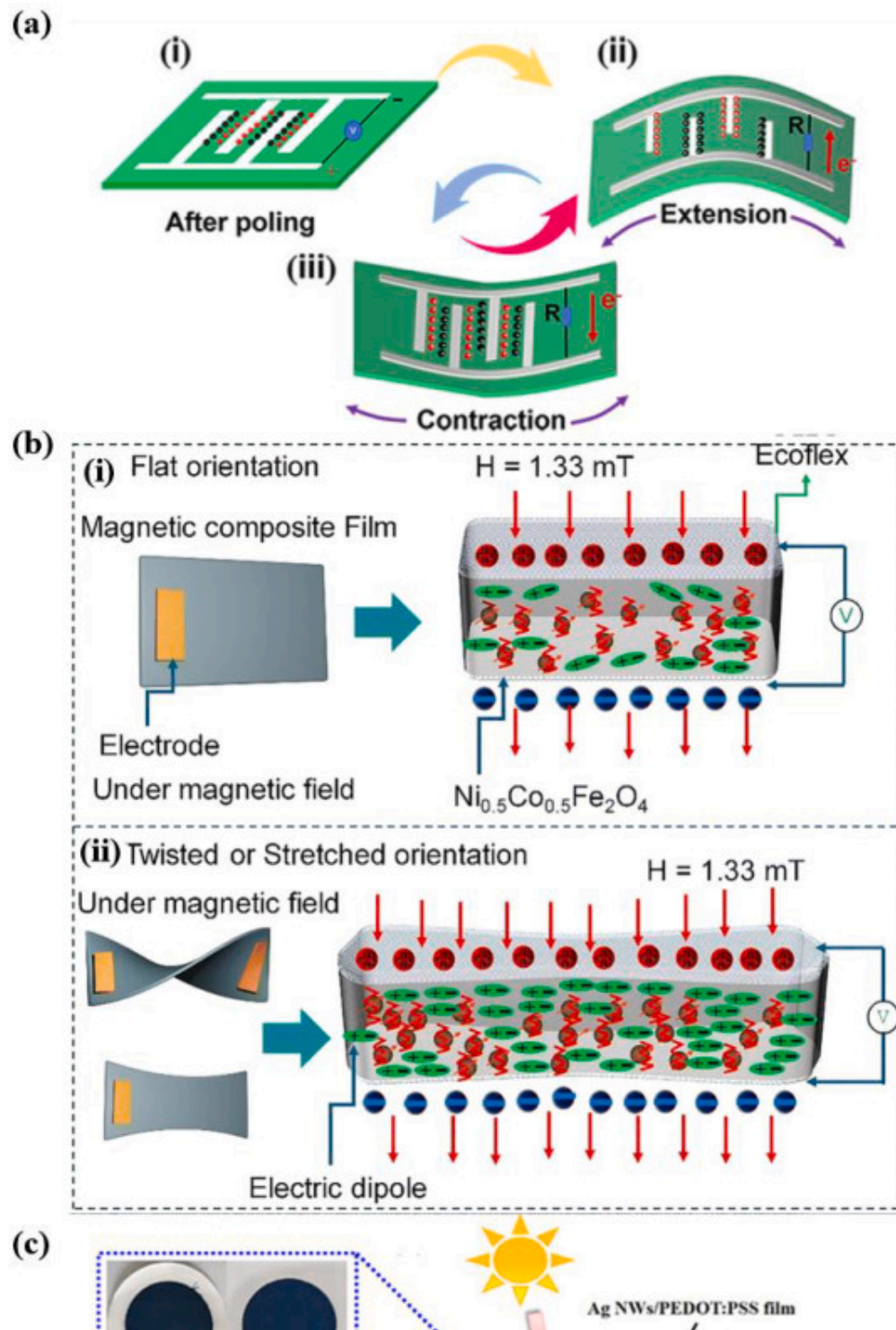
4.2.1. Mechano-electrical coupling

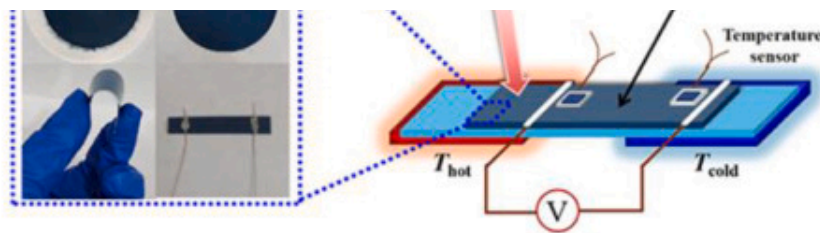
Mechano-electrical synergy addresses the intrinsic limitations of single-mode energy harvesters, specifically the low current density of the triboelectric effect (governed by surface contact electrification) and the low voltage and flexibility constraints of the piezoelectric effect (governed by internal dipole polarization). By integrating metals, ceramics, and polymers into a unified architecture, researchers can achieve a "Structure-Material Dual-Synergy." In this ternary setup, the polymer matrix serves as a highly deformable friction layer, the embedded piezoelectric ceramics provide strong bulk polarization, and the metallic or conductive 2D fillers act as localized charge-trapping

centers. This triadic interaction drastically reduces internal electrical impedance and enhances interfacial polarization, allowing the constructive superposition of triboelectric charges and piezoelectric potentials.

Shi et al. [104] successfully prepared a core-shell structured PMMA@BaTiO₃/NWs via surface-initiated in situ atom transfer radical polymerization (ATRP). The grafting of PMMA on the BaTiO₃ results in uniform dispersion of NWs in the PVDF-TrFE matrix and enhances the stress transfer at the interface between the piezoelectric NWs and the PVDF-TrFE matrix, leading to significantly enhanced output performance of PMMA@BaTiO₃/PVDF-TrFE based PENGs. The electrical output of such hybrid generators relies heavily on the ferroelectricity-coupled dielectric modulation within the polymer friction layer. Introducing a highly conductive phase alongside a piezoelectric ceramic can optimize these dielectric properties for maximum energy harvesting. For instance, Ghosh et al. [105] demonstrated a hierarchically engineered composite comprising barium titanate (BaTiO₃), 2D MXene, and thermoplastic polyurethane (TPU). In this specific system, the high-k ferroelectric ceramic induces strong local polarization under mechanical stress, while the highly conductive MXene nanosheets function as dispersed micro-electrodes throughout the polymer matrix. This configuration significantly increases the overall dielectric constant of the TPU composite while minimizing dielectric loss, yielding a massive enhancement in charge density that enables highly stretchable self-powered sensors for human biomechanics monitoring. Beyond mere compositional mixing, the geometric alignment of these ternary components plays a crucial role in maximizing the mechano-electrical conversion efficiency, particularly for tactile and acoustic sensing. Recent advancements by Deng et al. [106] and Lin et al. [107] have leveraged electrospinning technology to create structurally aligned nanocomposites utilizing BaTiO₃-incorporated PVDF-HFP nanofibers coupled with conductive networks like multi-walled carbon nanotubes (MWCNTs). The electrospinning process mechanically stretches the polymer, synergistically forcing the PVDF-HFP chains to transition into the highly electroactive beta-phase while simultaneously aligning the BaTiO₃ ceramic nanoparticles along the fiber axis (Fig. 10a). Under dynamic bending or

pressure, this aligned architecture efficiently channels mechanical stress to the ceramic fillers and maximizes the surface area for triboelectric charge generation, culminating in superior power density and ultra-fast response times for flexible tactile applications. Wang et al. [108] designed a wearable tactile sensor using Galinstan (Ga-In-Sn alloy) liquid metal as the sensing core. Integrated within a PDMS elastomer matrix, this device combines high conductivity ($>3.46 \times 10^6 \text{ S/m}$) and extreme deformability inherent to liquid metals with structural stability and conformal contact capability. Mounted on human fingers, the sensor enables simultaneous tactile/temperature monitoring during gripping tasks.





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Fig. 10. (a) Working principle of F-PENG [107], © 2023, American Chemical Society, (b) Illustration of magnetic–electric dipole interactions in ME films in (i) flat and (ii) twisted or stretched orientations. © 2026 Copyright Clearance Center, Inc. All rights reserved. [111]. (c) The image of the Ag NW/PEDOT:PSS film on the PVDF filter, PET substrate, and the slide-film sample and schematic diagram of the measurement setup for photothermoelectric performance testing [112]. © 2026 Rights managed by AIP Publishing.

Ultimately, the success of these mechano-electrical hybrid systems highlights the critical role of the conductive phase. While the polymer provides the necessary contact area and the ceramic contributes the dipole moment, the metallic or carbon-based network acts as a "Charge Bridge and Trap." By strategically tuning the concentration of these conductive fillers to remain slightly below the percolation threshold, designers can effectively prevent current leakage while maximizing interfacial capacitance. This delicate balance ensures that the generated mechanical energy is efficiently trapped and funneled into the external sensing circuit, achieving true self-powered, high-fidelity multimodal perception.

4.2.2. Magneto-electrical coupling

Detecting magnetic fields in soft wearable electronics necessitates an indirect, "strain-mediated" coupling mechanism, fundamentally translating magnetic stimuli into mechanical deformation and subsequently into electrical signals. Unlike rigid, single-phase multiferroics, ternary magnetoelectric (ME) composites rely on the synergistic

interaction between magnetostrictive fillers, piezoelectric ceramics, and a flexible polymer matrix. When exposed to an external magnetic field, the magnetostrictive components (such as FeSiB alloys or cobalt ferrite nanoparticles) undergo physical deformation. This mechanical strain is transferred across the heterogeneous interface through the elastomeric polymer network to the piezoelectric phase (such as P(VDF-TrFE) or BaTiO₃), which in turn generates a measurable electrical polarization. In this triadic architecture, the polymer matrix transcends its traditional role as a mere structural binder; it acts as the critical energy-transfer medium where "Interface Fidelity" and mechanical impedance matching determine the overall magnetoelectric conversion efficiency. To optimize this strain transfer and overcome the inherently low piezoelectric coefficient of pristine polymers, recent research has focused on hierarchically engineering the ternary interfaces. For example, a recent study demonstrated that doping the P(VDF-TrFE) polymer matrix with ceramic boron nitride (BN) nanoparticles significantly enhances the material's piezoelectric coefficient (d_{33}) up to 43 pC/N [109]. When this reinforced piezoelectric matrix was integrated with amorphous FeSiB magnetostrictive ribbons, the resulting flexible composite exhibited dramatically improved AC magnetic field sensing capabilities. The ceramic BN phase not only boosted the electrical output but also tuned the mechanical modulus of the polymer, minimizing energy dissipation during the magneto-mechanical energy transfer.

Beyond planar films, advancing the geometric and structural design of these composites has unlocked new paradigms in touchless human-machine interactions (HMI). Researchers have recently developed flexible magnetoelectric fibers and highly stretchable elastomeric magnetic films (utilizing matrices like Ecoflex embedded with Ni_{0.5}Co_{0.5}Fe₂O₄ nanoparticles) that conform seamlessly to the human skin [110,111]. By engineering micro-scale modulus gradients within the polymer matrix, these advanced architectures ensure that the magnetic strain is efficiently focused onto the active piezoelectric interfaces rather than being absorbed and damped by the soft elastomer (as shown in Fig. 10b). Consequently, these strain-engineered flexible sensors can harvest

ambient magnetic noise and detect low-intensity magnetic fluctuations induced by subtle finger movements, enabling real-time gesture recognition without physical contact.

The evolution of flexible magneto-electrical sensors underscores a vital design principle: the necessity of precise mechanical impedance matching. The polymer interface must be "stiff enough" to transmit the magnetostrictive strain with high fidelity, yet "soft enough" to maintain overall device conformability and prevent brittle fracture under large macroscopic deformations. Future innovations in ternary ME composites will likely hinge on such anisotropic structural engineering, enabling devices that simultaneously offer high magnetoelectric coefficients and extreme mechanical resilience for next-generation wearable spintronics and touchless control systems.

4.2.3. Photo-thermo-electrical coupling

Unlike direct photovoltaic energy harvesting, the photothermoelectric (PTE) coupling synergy relies on a sophisticated two-step cascade conversion: transforming incident light into localized heat, which is subsequently converted into a readable electrical signal. This intricate process exploits the complementary properties of ternary composites to overcome the narrow-band limitations of traditional semiconductor photodetectors. Within these hybrid architectures, metallic nanostructures or highly absorptive low-dimensional networks function as highly efficient "plasmonic heaters." By leveraging phenomena such as localized surface plasmon resonance or broad-spectrum photon trapping, these metallic phases absorb incident radiation—ranging from the ultraviolet to the deep terahertz regime—and rapidly dissipate it as localized thermal energy. This robust photothermal effect establishes a steep temperature gradient across the active interface. Subsequently, the thermoelectric phase, typically comprising high-ZT semiconductor ceramics or intrinsically conductive polymers, translates this newly formed thermal gradient into a measurable voltage via the Seebeck effect. Throughout this entire cascade, the flexible polymer matrix plays a dual and seemingly contradictory role. It must act as a resilient structural binder that maintains excellent electrical percolation for charge transport, while simultaneously serving as an effective thermal

barrier (often via porous or aerogel designs) to prevent rapid heat dissipation and preserve the critical temperature difference.

Recent scientific advancements have prominently validated this "Light-Heat-Electricity" triadic synergy, shifting the research paradigm from simple material blending to engineered interface optimization. For instance, recent studies in 2025 have demonstrated that integrating silver nanowires (Ag NWs) into a conductive PEDOT:PSS organic matrix dramatically enhances overall photothermoelectric energy conversion (Fig. 10c) [112]. In this system, the metallic nanowires exponentially increase broadband light absorption, generating a maximum internal temperature difference nearly twice that of pristine polymer films under standard solar illumination, thereby significantly amplifying the output voltage for self-powered sensing. Similarly, incorporating multi-walled carbon nanotubes (MWCNTs) or 2D MXene nanosheets into thermoelectric polymer matrices has facilitated the fabrication of ultra-flexible, room-temperature broadband infrared detectors with extraordinary sensitivity and flexibility [113]. These advanced ternary systems successfully decouple the optical absorption function from the thermoelectric generation process, empowering researchers to independently maximize both physical parameters. Ultimately, the realization of high-performance PTE sensing hinges on the fundamental principle of "Thermal-Electrical Impedance Management." The structural design must elegantly resolve the inherent conflict between phonon blocking and electron transmitting, ensuring that the photogenerated heat remains strictly localized at the active thermoelectric junctions without impeding the rapid extraction of diagnostic electrical signals.

4.3. Multimodal signal processing and decoupling strategies

4.3.1. Spatial and structural decoupling

The most fundamental and direct approach to mitigating cross-sensitivity in multimodal sensors relies on physical spatial isolation and structural decoupling, ensuring that distinct external stimuli are channeled through independent transduction pathways

before they can interfere. By abandoning simple homogeneous blending in favor of rationally designed heterogeneous architectures, ternary metal-ceramic-polymer composites can achieve true signal orthogonality.

A prominent example of this strategy is the development of asymmetric Janus and core-shell architectures. By engineering a Janus thin-film layout, researchers can physically isolate different sensing mechanisms—such as piezoelectric and triboelectric effects—on opposite faces of a single flexible substrate [114]. This asymmetric configuration allows a single device to simultaneously process distinct physical interactions without signal overlap, maintaining an ultra-compact footprint. Similarly, in core-shell microstructures, a conductive metallic or carbon-based core can be exclusively dedicated to piezoresistive strain sensing, while a specialized ceramic or metal-organic framework (MOF) shell selectively interacts with environmental factors like gas molecules or humidity. The structural boundary inherently prevents electron leakage and isolates the piezoresistive network from chemical-induced baseline drift, guaranteeing high signal fidelity.

Furthermore, macroscopic vertical stratification and sandwich structures offer a highly effective solution for decoupling thermal and mechanical signals, which notoriously suffer from severe cross-talk. By utilizing the layer-by-layer assembly capabilities of polymeric materials, sensors can be constructed with an integrated thermal barrier—such as a porous polymer aerogel—sandwiched between a top piezoresistive pressure-sensing layer and a bottom thermoelectric temperature-sensing layer [115]. This vertical isolation ensures that transient environmental heat does not diffuse instantaneously into the pressure-sensitive conductive network, physically uncoupling thermal interference from the mechanical response. Furthermore, to reconstruct complex tactile information such as simultaneous normal and shear forces, modern sensor designs employ in-plane spatial isolation via high-density anisotropic sensor arrays [116]. By geometrically arranging distinct micro-sensing units across a flexible printed circuit substrate, the vector components of multidimensional forces are mechanically separated at the structural level before their conversion into electrical signals. Ultimately, this hardware-level spatial decoupling not only minimizes hysteresis but also significantly reduces the

computational burden required for subsequent data processing, establishing a robust physical foundation for advanced electronic skins and intelligent human-machine interfaces.

4.3.2. Frequency and temporal domain decoupling

When structural isolation is restricted by the necessity for extreme device miniaturization or seamless tissue integration, researchers must extract multiple independent parameters from a single, homogeneously blended sensing layer. This challenge necessitates shifting the decoupling paradigm from the spatial domain to the frequency and temporal domains. Instead of viewing a ternary metal-ceramic-polymer composite as a simple variable resistor, this advanced strategy treats the material as a complex electrical network comprising interconnected capacitive and resistive elements. These internal components respond fundamentally differently to external stimuli under varying alternating current (AC) frequencies or transient time frames, providing a mathematical basis for signal separation.

In the frequency domain, the decoupling mechanism elegantly exploits the fact that distinct physical phenomena exhibit divergent frequency dependencies. For instance, in flexible ion-conducting polymer composites doped with conductive nanoparticles, mechanical pressure typically alters the macroscopic contact area or the internal tunneling distance. This structural deformation results in a geometric capacitance change that remains relatively stable across a broad frequency spectrum. Conversely, temperature variations predominantly influence the intrinsic mobility of charge carriers, inducing conductivity fluctuations that are highly sensitive to the applied AC frequency. Recent technological advancements have successfully capitalized on this intrinsic divergence. By utilizing multifunctional polymer matrices (such as chitosan or ionic hydrogels), researchers have engineered dual-modal sensors where pressure and temperature responses are mathematically isolated simply by modulating the input AC frequency and analyzing the resulting impedance spectra [117]. This frequency-enabled approach empowers a single, ultra-thin film to output decoupled thermal and mechanical readings

with exceptional accuracy (>92%), drastically simplifying the physical architecture of electronic skins.

Parallel to frequency modulation, temporal domain decoupling leverages the transient relaxation dynamics of the composite materials following a sudden electrical or physical excitation. Viscoelastic polymers and ternary ionogels inherently possess time-dependent electrical behaviors that can be meticulously analyzed to separate overlapping signals. A recent breakthrough in this area involves tracking the "charge relaxation time"—the specific duration required for ions or electrons to re-establish equilibrium within the polymer matrix after a perturbation. Recent 2024 studies on multimodal soft sensors, such as those utilizing ionic liquids embedded in poly(ethylene glycol) gels (PEGgel), have definitively demonstrated that mechanical stretching induces an instantaneous change in steady-state capacitance but leaves the fundamental ion relaxation dynamics completely unaffected [118]. In stark contrast, ambient temperature fluctuations exponentially alter the charge relaxation time due to localized changes in matrix viscosity and thermally activated ion hopping rates. By applying a brief step-voltage pulse and simultaneously measuring both the immediate capacitive magnitude and the subsequent exponential decay constant in the time domain, the system can independently quantify strain and temperature without any cross-talk. Ultimately, these frequency and temporal strategies elegantly transfer the burden of signal decoupling from bulky hardware modifications to sophisticated read-out electronics, paving the way for high-density, interference-free multimodal perception.

4.3.3. Data-driven and machine learning decoupling

When physical isolation and frequency modulation reach their fundamental limits within highly miniaturized or ultra-conformal ternary composites, researchers increasingly turn to data-driven and machine learning (ML) architectures to achieve robust signal decoupling. This paradigm shift fundamentally transitions the decoupling burden from intricate hardware engineering to sophisticated software algorithms, offering a powerful "soft" solution for highly coupled, non-linear sensing environments. Instead of struggling

to completely eliminate cross-talk at the material interface, this strategy deliberately embraces the complex, superimposed responses generated by the metal-ceramic-polymer network. Advanced deep learning models, particularly Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), are deployed to process the raw, multidimensional electrical matrices output by the sensors. Because divergent physical stimuli—such as temperature, humidity, and mechanical strain—induce distinct temporal and spatial "fingerprints" in the sensor's electrical response, these advanced algorithms can automatically extract high-dimensional hidden features that traditional analytical methods fail to resolve.

Recent scientific breakthroughs vividly illustrate the transformative potential of artificial intelligence in flexible tactile perception. For instance, pioneering research published in 2025 has successfully integrated deep learning-assisted classification frameworks with multisensory electronic skins to achieve interference-free, real-time decoupling of pressure and temperature [119]. By feeding the highly coupled piezoresistive and thermoelectric data into a customized neural network, the system can effortlessly distinguish the complex, experience-driven features of grasped objects, evaluating their simultaneous softness (modulus) and warmth (thermal conductivity) with unprecedented classification accuracies exceeding 98% [120]. Furthermore, when these ternary composites are fabricated into high-density sensory arrays, the resulting data matrix unlocks the potential for advanced "Information Fusion." The machine learning model actively cross-references the subtle signal variations across adjacent sensing pixels, mathematically canceling out environmental drift and precisely isolating the target variables. As highlighted in comprehensive recent reviews on AI-enhanced electronic skins [121], the future of multimodal decoupling lies undeniably in "Hardware-Algorithm Co-Design." In this emerging approach, materials scientists intentionally engineer the metal-ceramic-polymer interfaces not merely for absolute sensitivity, but to generate sufficiently complex, information-rich dynamic signals that are optimized specifically for algorithmic decoding.

5. Key applications

5.1. Gas/liquid sensing

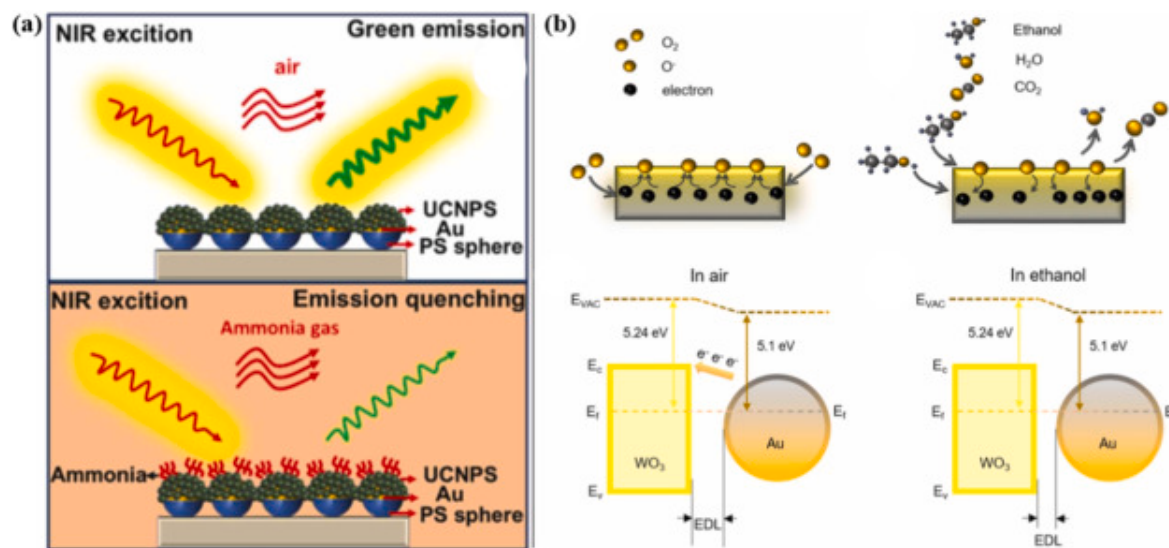
The application of metal/ceramic/polymer composites in gas/liquid sensing is mainly due to their multi-component synergistic effect and multi-scale structural design providing innovative solutions for highly sensitive and selective detection in complex liquid environments. Polymer-matrix composites differ by their frequent application in gas sensing, especially in the detection of hazardous gases and environmental monitoring, due to their flexibility, low cost and tunable electrical properties. In contrast, ceramic-based variants dominate high-temperature environments, such as automotive emission monitoring and industrial gas detection due to their excellent thermal stability and corrosion resistance [122].

Meanwhile, metal-matrix composites excel in liquid sensing domains in liquid sensors due to their excellent mechanical strength and electrical conductivity, especially in the detection of changes in pressure, temperature and liquid composition [123]. By compositing these materials, it is possible to not only optimize the response time and sensitivity of the sensors, but also improve their stability and durability for a wide range of extreme environments. By organically combining the conductive network of metals, the chemical stability of ceramics, and the molecular recognition function of polymers, composites are able to achieve multi-parameter sensing of liquid composition, concentration, viscosity, and ionic strength. Common gas sensing applications include the detection of toxic gases (e.g., CO [124], Nox [125], H₂S, [[126], [127], [128]]) and temperature changes.

The surface properties of the composites, such as surface activity, adsorption capacity and chemical reactivity, can also be used to further optimize the performance of the sensors by adjusting the ratio of the different components. In the detection of ethanol [129], constructed PTFE/Au/WO₃ heterostructures via sequential ion-beam/RF-sputtering. The sensor exhibited enhanced ethanol detection with high sensitivity (>85% @100ppm), fast

response/recovery (<15s), wide linear range (5-500ppm), superior selectivity and 30-day stability. Chen et al. [130] developed a UCNPs/Au@PS heterostructure for NH₃ sensing. Gold's plasmonic enhancement and interfacial stability enable cost-effective detection with rapid response (<8s), high SNR (>42dB), and 90-days signal stability.

To enhance the sensing performance, Zhu et al. [131] synthesized novel sea cucumber-shaped polyaniline/iron oxide (PANI/Fe₂O₃) nanocomposites using a simple and efficient one-step hydrothermal process, and the nanocomposites were further assembled onto polyethylene terephthalate (PET) flexible substrates, and found that the two sensors exhibited high selectivity for NH₃, and that the PANI/Fe₂O₃ nanocomposite-based sensors had good response and low detection limits (20±5ppm) at room temperature (0.3°C). Foronda et al. [132] advanced polyaniline/zinc oxide (PANI/ZnO) composite gas sensors for multi-VOC detection (e.g., ammonia, acetone, formaldehyde, methanol, ethanol). These sensors exhibit tunable room-temperature conductivity with characteristic saturation behavior. Notably, such nanostructured composites have emerged as a research focus due to their enlarged specific surface area and intensified gas-molecule interactions (Fig. 11), substantially improving sensing performance.



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Fig. 11. (a) Schematic representation of ammonia molecules adsorbed onto the surface of upconversion nanoparticles (UCNPs) [130] © 2025 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies, (b) Schematic representation of the ethanol sensing mechanism of the PTFE/Au/WO₃ sensor [129] © 2025 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

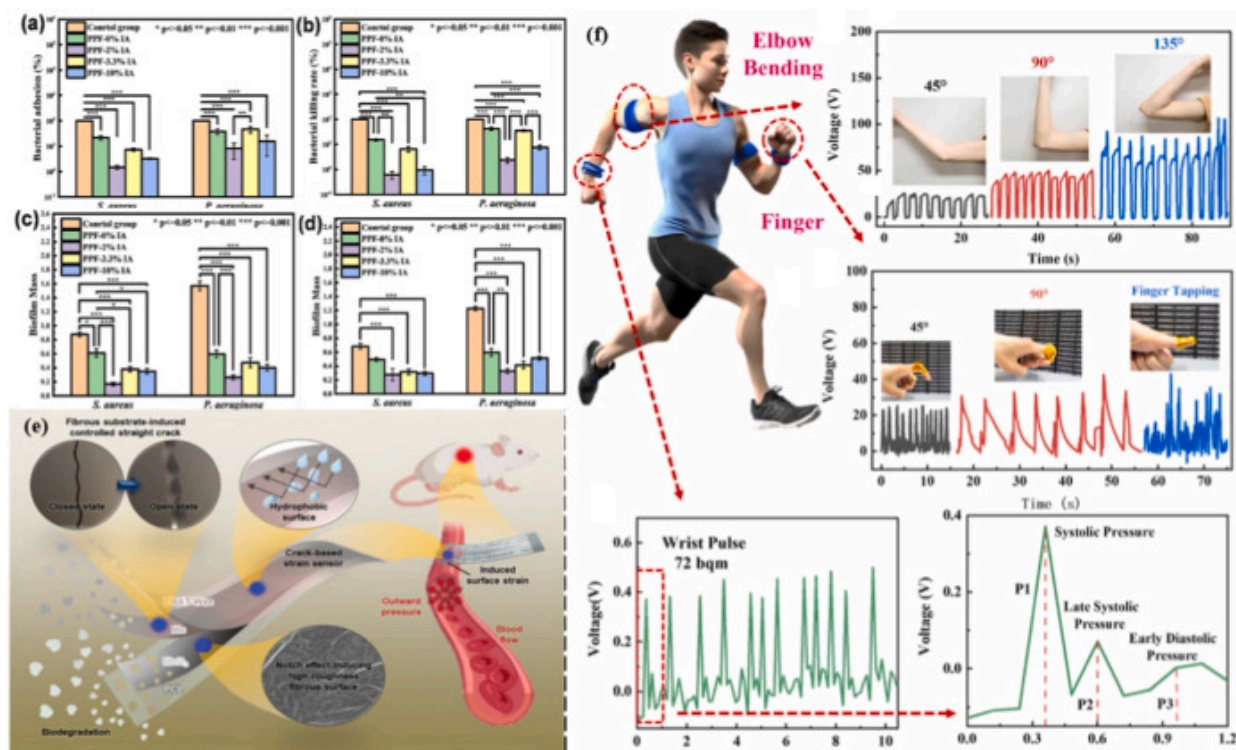
5.2. Human health

The use of metal/ceramic/polymer composites in human health monitoring is gradually increasing, especially in the field of wearable devices and biosensors. These sensors are integrated into wearable devices such as smart watches. They are capable of monitoring multiple healthcare parameters simultaneously [133]. These wearable sensors enable continuous, real-time tracking of vital signs and physical activity. Their integration into healthcare is revolutionizing preventive strategies, disease management, and personalized health monitoring [134]. Key attributes include adaptability, flexibility, and notably, non-invasiveness [135]. This latter feature eliminates the need for surgical implantation or invasive procedures, representing a significant advantage. Facilitating unobtrusive and ongoing health surveillance during daily activities, these sensors integrate real-time tracking of vital signs and movement. Crucially, the synergy between their non-invasive attributes and advanced data processing optimizes preventive care strategies and health management outcomes [136]. In addition, some composites can be used to detect physiological parameters such as blood glucose, respiration [[137], [138], [139]], sweat composition and other physiological parameters for real-time monitoring of human health status.

Polymer materials are commonly used to make biosensors that come into contact with the skin due to their good flexibility and malleability. Ceramic materials, on the other hand, are widely used in implantable medical devices due to their high biocompatibility and stability [140], but the brittleness of ceramics can be a drawback for applications such

as flexible smart devices, and composites of polymers with ceramics have been widely investigated [141,142]. Bioactive capabilities have been identified in select polymer/ceramic composites. Polymer composites, meanwhile, enable versatile bone repair strategies through machinable customization, low-temperature sintering compatibility, and tissue scaffold fabrication (e.g., via electrostatic spinning technology) [143].

In addition, polymer composites can be customised to have more bone-like thermal properties than metallic materials. Metallic materials in particular have high electronic conductivity in electronic sensors and are suitable for high precision health monitoring systems [144]. In order to address the shortcomings of electrochemical detection sensors with low sensitivity, narrow response range and high detection limit, most of the existing studies have been based on the introduction of nanostructured metal oxides in the sensors by polymers [[145], [146], [147]]. Xiong et al. [148] fabricated PVDF nanofibers (22 ± 15 nm) with homogeneous BN nanoparticle doping, attaining 99.1% β -phase content (Fig. 12). These monolayer (macroscopic) through multilayer (micro) composites demonstrate high sensitivity coupled with self-powered functionality, suitable for diverse applications including: laryngeal vibration detection, human activity monitoring, and alarm-system microswitches. Zhou et al. [149] engineered a flexible capacitive sensor containing 60vol% liquid metal (LM), achieved via KH580 silane coupling agent self-assembly through electrostatic interactions and HPDMS crosslinking. This high-LM-loading device demonstrates robust real-time monitoring fidelity for minute motion artifacts, sound vibrations, and peripheral pulse dynamics. The combination of composites can ensure the performance of the sensors while improving their comfort, durability, and long-term stability, opening up possibilities for remote health monitoring and smart medicine.



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Fig. 12. Antimicrobial properties of PPF-IA hydrogel: (a) Inhibition of adhesion, (b) Bacterial killing rate, (c) Inhibition of biofilm formation, (d) Disruption of biofilm of hydrogel [144] © 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies. (e) Schematic of a fully biodegradable crack-based strain sensor with controllable crack shape and lifetime for biomechanical signal monitoring [151] © 2024 The Author(s). Advanced Functional Materials published by Wiley-VCH GmbH. (f) Schematic of an 8-layer MPP-2 sensor for monitoring elbow flexion at different angles, finger movement and and wrist pulse [148] © 2024 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

Deep learning's data-driven processing—characterized by ultra-efficient pattern recognition—is fundamentally transforming societal infrastructures. Exemplifying this shift, Yu's team, Yu et al. [150] fabricated directionally-aligned SCNM foam sensors (SA/CHI/nanofibre/MPS) using electrostatic spinning coupled with directional lyophilization. The resulting wearable platform performs cloud-connected physiological monitoring, facilitating dynamic motion analysis and multi-user remote diagnostics. Li et al. [151] presented a fully biodegradable, ultra-sensitive, crack-based, flexible strain sensor by integrating a molybdenum (Mo) membrane, a molybdenum trioxide (MoO_3 =GF) at 1.5% strain for 1355) adhesion layer and polycaprolactone (PCL) substrate, the sensor shows great potential for highly sensitive sensors by short-term (≈ 3 days) monitoring of vascular pulsations demonstrating the potential of sensing tools for rapid, accurate, and time-dependent disease diagnosis and treatment. Fig. 14 lists the applications of composite sensors in human health.

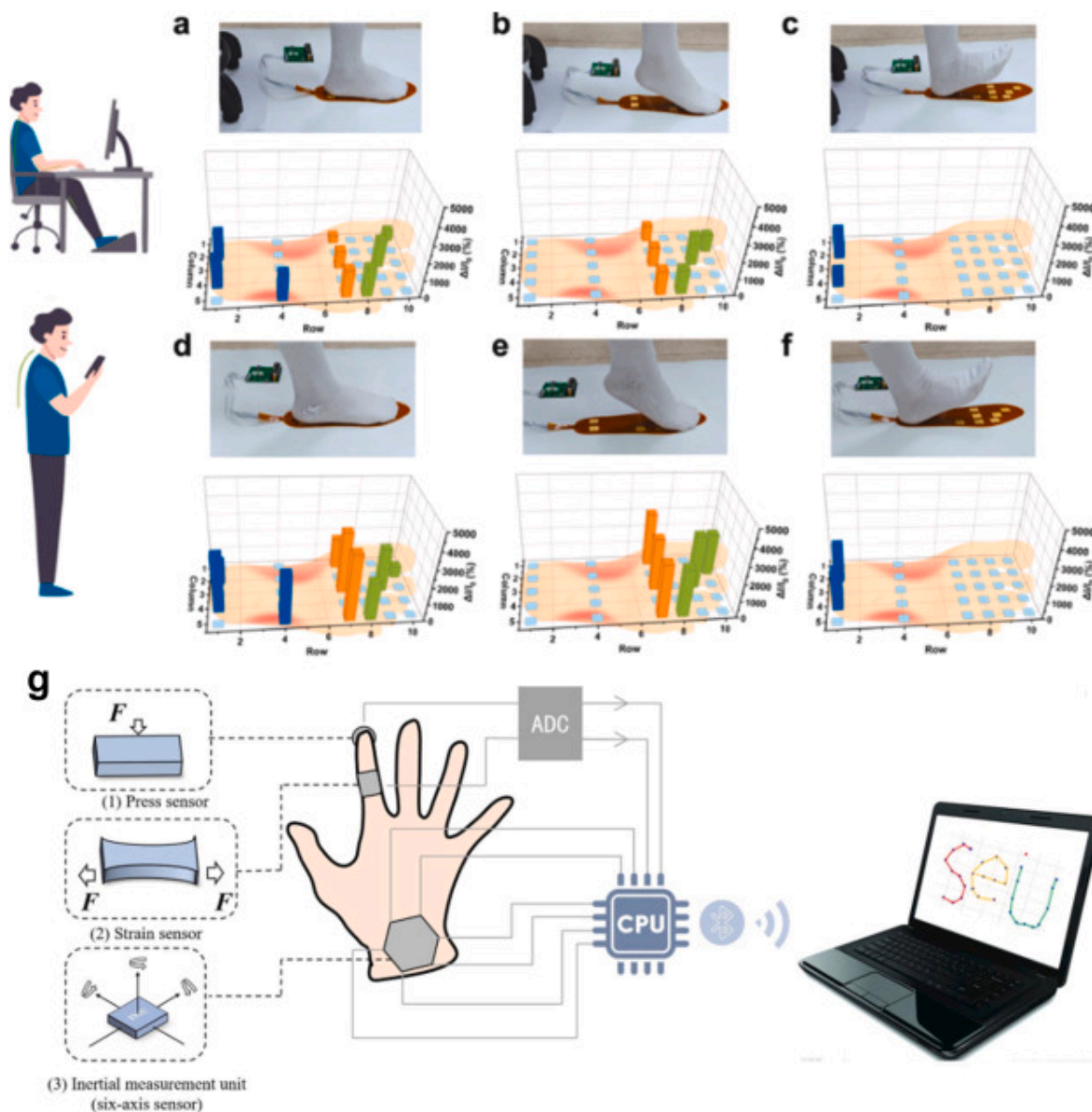
5.3. Human-computer interaction

Rapid global demographic ageing intensifies requirements in three domains: geriatric health surveillance, assisted living support, and risk prevention. Longitudinal monitoring of biosignals and kinetic data constitutes a foundational approach to preemptive healthcare intervention, injury avoidance, and quality-of-life optimization [152]. Wearable sensors for physiological state monitoring, motion recognition and human-computer interaction have shown great potential. However, traditional rigid devices (e.g., clip-on optical pulse sensors, inertial measurement units (IMUs), smartwatches) face comfort and power limitations that hinder long-term geriatric care applications [[153], [154], [155]]. The development of flexible sensors with excellent wearer comfort offers new solutions to this challenge.

Applications in the field of human-computer interaction are mainly in the areas of haptic feedback, sensor interfaces and intelligent systems. Composite materials can be used to develop smart gloves, virtual reality devices, and touch interfaces by integrating electronic components and sensors. These materials enable precise motion sensing and

environmental response, enhancing the immersion and sensitivity of human-computer interaction. Polymer composites are commonly used in touch screens and haptic sensors due to their flexibility and responsiveness [156], while metallic materials are commonly used for driving and transmitting signals, providing stable conductivity [157].

Ceramic materials, on the other hand, can be used to improve the thermal stability and mechanical strength of the system. By adjusting the composition and structure of composites, multifunctional devices with better interaction experience can be obtained, which has a wide range of applications in the fields of smart devices, virtual reality, and robotics [[158], [159], [160]], as shown in Fig. 13, Liu et al. [44] reported the design of a bicontinuous structure that can be fabricated by doping a thermoplastic polyurethane (TPU) framework with liquid metal (LM) for highly stretchable fabrication. The LM phase and the TPU frame interpenetrate with each other to form a high-speed heat conduction channel and provide a strong backbone for stress transfer, paving the way for the development of highly stretchable materials for thermal management and opening up new possibilities for future wearable electronics.

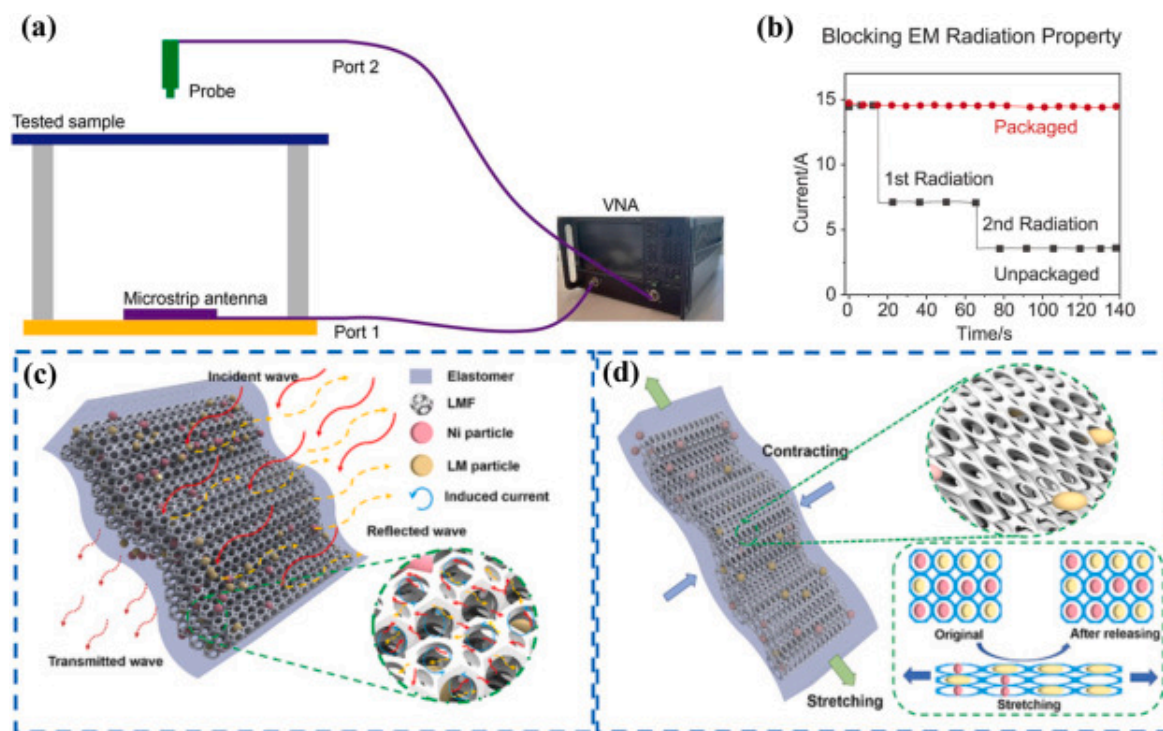


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Fig. 13. Response of the SBS-TA-7%/LM plantar pressure sensor during (a) sitting, (b) heel-on-ground, (c) forefoot-on-ground, (d) standing, (e) heel-on-ground standing, and (f)

forefoot-on-ground standing [147] © 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies. (g) Schematic diagram of the components in the HCI glove [161] © 2025 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.



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Fig. 14. (a) Schematic diagram of the NF shielding test system. EM signals are transmitted by the microstrip antenna and received by the probe. (b) EM shielding performance of LMP [169] © 2024 the authors, some rights reserved; exclusive licensee American Association for the Advancement of Science. Graphical representation of EMI shielding mechanism of LMF-EC under (c) free stress and (d) tension [173] © 2022 Elsevier B.V. All rights reserved.

While critically relevant to flexible systems, stretchable liquid-metal devices remain hampered by processing complexities and suboptimal interfacial adhesion kinetics at the metal-polymer interface. Zheng et al. [162] reported that electrostatically spun polymer fibre membranes containing semi-embedded liquid metal particles can be used for the fabrication of stretchable electronic products. film can be used to fabricate stretchable electronics. The liquid metal particles within the fibre network break up under pressure and fill the gaps in the fibre network to form conductive regions. This allows use of circuit pattern stamps to create circuits with high resolution (minimum line width of 50 μ m) and stability (more than 30,000 cycles at 100% strain). The Ag-TPU-based wearable virtual mouse developed by Wang's team [161] establishes a novel paradigm for coupling flexible materials with interactive systems. It extends multi-modal sensing capabilities while ensuring temporally-precise information transfer in HCI processes.

Escalating demand for flexible-stretchable electronics—driven by bio-integrated wearables requiring seamless human-body conformability—has accelerated research into conductive composite systems exhibiting synergistic flexibility, mechanical resilience, and electrical performance. Kim et al. [163] engineered high-reliability 3D nanocomposites through directional freeze-drying, utilizing PDMS soft substrates and PVA/AgNW rigid cores. AgNW reinforcement enabled ultra-strong cores, with PDMS infiltration arresting: (i) micro-crack advancement and (ii) lyophilized framework collapse at extreme compression—critical for flexible device durability. Leveraging mussel-inspired adhesion, Liu et al. [164] engineered electrospun PU membranes with bimetallic conductive layers (AgNPs/CNTs). This strain sensor achieved whole-body biomechanical monitoring—from subtle physiological vibrations to macro-joint kinematics—alongside robust human-computer interaction capabilities.

5.4. Electromagnetic shielding

With the rapid development of electronic industry, electromagnetic pollution has seriously affected people's daily life [165]. Electromagnetic interference (EMI) and radiation not only interfere with the normal operation of electronic equipment, but also

threaten the information security of communication equipment, and even endanger human health [166]. Therefore, the need for various aspects of EMI shielding materials is becoming more and more urgent. The interference of electromagnetic radiation on electronic devices may lead to signal distortion or equipment failure, thus effective EMI shielding materials become critical in the design of electronic devices.

Metal composites are excellent in shielding high-frequency electromagnetic waves due to their good electrical conductivity; ceramic materials have an advantage in shielding low-frequency electromagnetic waves, and at the same time, they have good high-temperature resistance; polymer composites are able to combine the advantages of other materials, which can not only provide effective shielding performance, but also reduce weight and increase flexibility. Polymer-based conductive adhesives with embedded fillers exhibit superior flow dynamics and gap-filling capacity [167,168], enabling direct EMI shielding and thermal conduction in micro-scale channels. Nevertheless, mandatory dielectric isolation layers to prevent electrical shorts increase packaging complexity and impede electronics miniaturization [169]. A wide range of shielding against electromagnetic waves of different frequencies can be achieved by designing multilayer structures of composite materials [62,64,170,171].

As shown in Fig. 14, Yao et al. [172] reported that the EMI SE of three-dimensional (3D) LM skeleton matrix polymer composites increased to more than twice the EMI SE at 400% strain, which reached the shielding level of metal plates measured under the same conditions. Yu et al. [173] developed a super-stretchable liquid metal-foam-elastomeric composite (LMF-EC), capable of enduring 800% strain. This material exhibited exceptional electrical conductivity ($\approx 10^4 \text{ Scm}^{-1}$) and thermal conductivity ($17.6 \text{ Wm}^{-1} \text{ K}^{-1}$). Notably, it also demonstrated an unexpected stretch-enhanced EMI shielding performance of 85 dB, attributed to the elongation and reorientation of the conductive network under strain.

Owing to reversible hydrophobic associations and hydrogen bonding interactions, the hydrogel composite boasts excellent EMI shielding effectiveness (EMI SE) as well as self-healing ability, and can still stably monitor human movements as a sensor after being

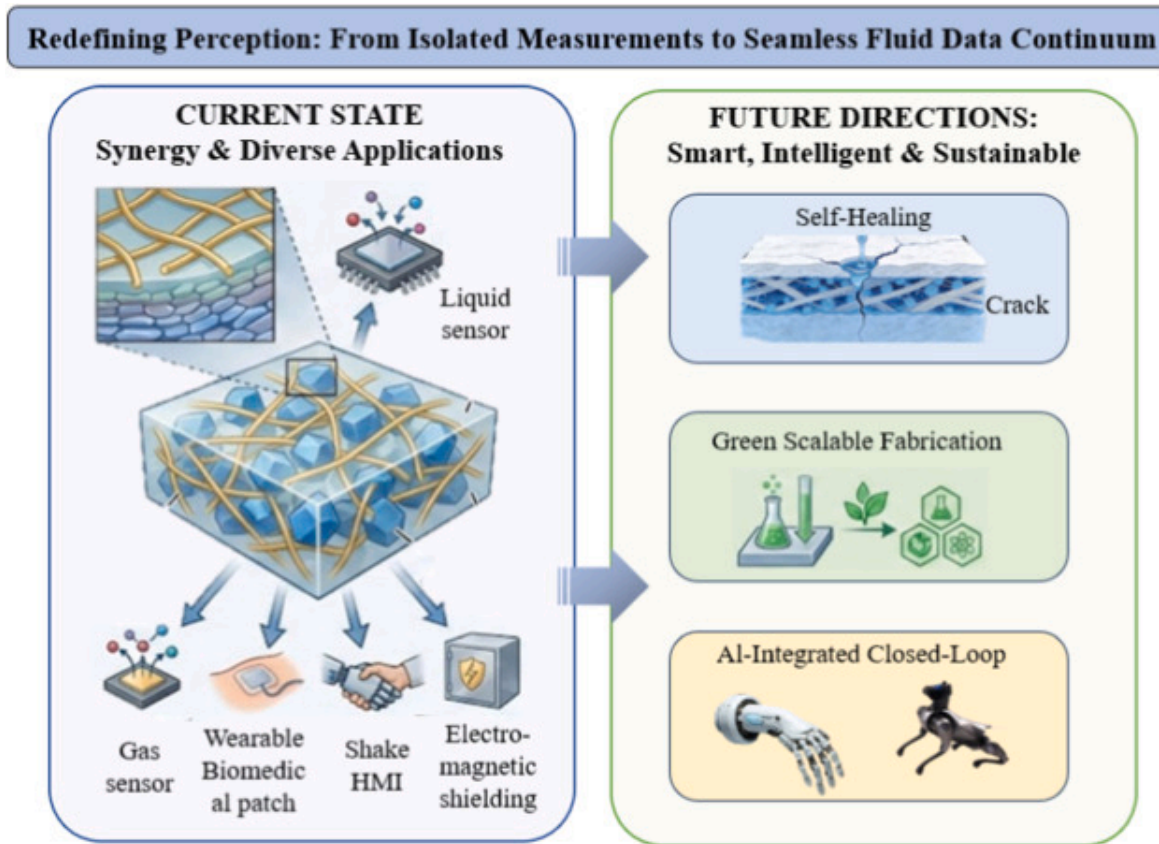
healed. Huang et al. [174] documented the preparation of superserviceable hydrogels through in-situ polymerization of acrylamide (AM) and N-acryloyl-11-aminoundecanoic acid (A-11) via directional freezing within pre-fabricated AgNWs aerogels. The AgNWs-based conductive network formed in the composites endows the hydrogel composites with high electrical conductivity, enabling them to exhibit superior resistive-strain responses under ultrahigh tensile strains. Additionally, thanks to the three-dimensional (3D) conductive honeycomb structure, the hydrogel composites also show exceptional EMI shielding performance. With the rapid development of smart devices and communication technologies, these composites play a crucial role in the design of EMC in automotive, aerospace and communication equipment.

6. Conclusion and outlook

This review has systematically elucidated the electro-mechanical and photo-chemo-electrical coupling effects at ternary metal-ceramic-polymer interfaces, transitioning the field's focus from empirical material blending to mechanism-driven rational design. By classifying synergistic effects into interfacial charge transfer, stress transduction, and multi-physics coupling, we have clarified how conductive networks, piezoelectric ceramics, and flexible polymers can be engineered to achieve signal fidelity far beyond simple superposition. Crucially, through a comprehensive quantitative analysis of cross-study data, we have established the fundamental structure-performance regulation rules, identifying the critical percolation thresholds, interfacial binding forces, and network connectivity required to balance sensitivity with mechanical robustness. Furthermore, we dissected the cutting-edge strategies for multimodal signal decoupling, demonstrating how spatial architecture, frequency-domain modulation, and data-driven algorithms can effectively resolve the inherent cross-talk in complex sensing environments.

Despite these transformative advances, the path toward fully autonomous electronic skins grapples with persistent trade-offs between high sensitivity, multimodal selectivity, and long-term environmental adaptability. Future breakthroughs will likely hinge on three convergent directions (Fig. 15):

- (1) From empirical trial-and-error to quantitative rational co-design:
Leveraging the quantitative structure-performance laws established in this review, future studies should integrate materials informatics with finite element modeling to realize precise prediction of composite sensing performance. This approach enables pre-optimization of gradient microstructural length scales and interfacial energy band alignments at the material design stage, laying a theoretical foundation for targeted fabrication of high-performance composites.
- (2) Hardware-Algorithm Synergistic Decoupling: The seamless integration of data-driven analysis is no longer optional but foundational. The next generation of sensors must evolve into "Hardware-Algorithm Co-designed" systems, where the material's intrinsic physico-chemical response features (e.g., transient relaxation times, complex impedance spectra) are intentionally engineered to be decidable by machine learning models, enabling real-time, drift-free perception.
- (3) Intrinsic functional modification for adaptive and sustainable composite design: Relying on material chemistry innovation, the development of intrinsic self-healing chemistries for metal-ceramic-polymer interfaces and green, scalable fabrication processes (e.g., template-guided assembly) will be prioritized. This will enhance the environmental adaptability and long-term stability of composite sensors under extreme service conditions, while minimizing the ecological footprint of material preparation and application, and bridging the gap between laboratory prototypes and practical sensing applications.



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Fig. 15. Evolutionary roadmap of metal-ceramic-polymer composites.

Future research rooted in material chemistry innovation—focusing on interfacial engineering, microstructural regulation, and functional modification—will continue to break through the performance trade-offs of composite sensors, and further promote the development and practical application of advanced sensing science and technology. The integration of these material design principles with sensing system engineering will drive the creation of next-generation high-sensitivity, selective, and stable multi-modal

sensors, opening up new application scenarios in precision detection, intelligent sensing, and other related fields.

CRediT authorship contribution statement

Meixuan Peng: Writing – review & editing, Writing – original draft, Visualization, Conceptualization. **Qiong Wu:** Writing – original draft, Resources. **Xin Zhao:** Visualization, Resources. **Xiaoyuan Zhang:** Writing – review & editing, Writing – original draft, Supervision, Funding acquisition. **Zheng Sun:** Writing – original draft, Funding acquisition. **Zhiqiang Su:** Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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